

Movement Patterns of the Knee During Gait Following ACL Reconstruction: A Systematic Review and Meta-Analysis

Mandeep Kaur¹ · Daniel Cury Ribeiro¹ · Jean-Claude Theis² · Kate E. Webster³ · Gisela Sole¹

Published online: 2 March 2016
© Springer International Publishing Switzerland 2016

Abstract

Background Altered gait patterns following anterior cruciate ligament reconstruction (ACLR) may be associated with long-term impairments and post-traumatic osteoarthritis.

Objective This systematic review and meta-analysis compared lower limb kinematics and kinetics of the ACL reconstructed knee with (1) the contralateral limb and (2) healthy age-matched participants during walking, stair climbing, and running. The secondary aim was to describe the differences over time following ACLR for these biomechanical variables.

Method Database searches were conducted from inception to July 2014 and updated in August 2015 for studies exploring peak knee angles and moments following ACLR during walking, stair negotiation, and running. Risk of bias was assessed with a modified Downs and Black quality

index for all included studies, and meta-analyses were performed. Forest plots were explored qualitatively for recovery of gait variables over time after surgery.

Results A total of 40 studies were included in the review; 26 of these were rated as low risk and 14 as high risk of bias. The meta-analysis included 27 studies. Strong to moderate evidence indicated no significant difference in peak flexion angles between ACLR and control groups during walking and stair ascent. Strong evidence was found for lower peak flexion moments in participants with ACLR compared with control groups and contralateral limb during walking and stair activities. Strong to moderate evidence was found for lower peak adduction moment in ACLR participants for the injured compared with the contralateral limbs during walking and stair descent. The qualitative assessment for recovery over time indicated a pattern towards restoration of peak knee flexion angle with increasing time from post-surgery. Peak knee adduction moments were lower within the first year following surgery and higher than controls during later phases (5 years).

Conclusion Joint kinematics are restored, on average, 6 years following reconstruction, while knee external flexion moments remain lower than controls. Knee adduction moments are lower during early phases following reconstruction, but are higher than controls, on average, 5 years post-surgery. Findings indicate that knee function is not fully restored following reconstruction, and long-term maintenance programs may be needed.

Electronic supplementary material The online version of this article (doi:10.1007/s40279-016-0510-4) contains supplementary material, which is available to authorized users.

✉ Gisela Sole
Gisela.sole@otago.ac.nz

¹ Centre for Health, Activity and Rehabilitation Research, School of Physiotherapy, University of Otago, Box 56, Dunedin 9054, New Zealand

² Department of Surgical Sciences, Medical School, University of Otago, Dunedin, New Zealand

³ School of Allied Health, La Trobe University, Melbourne, VIC, Australia

Key Points

This systematic review and meta-analysis compared lower limb kinematics and kinetics of the anterior cruciate ligament reconstructed (ACLR) knee with (1) the contralateral limb and (2) healthy age-matched participants during walking, stair climbing, and running.

Knee kinematics are likely to be similar in ACLR participants at 6 years following surgery compared with healthy controls. Knee peak adduction moments are lower in ACLR participants 3.5 years following reconstruction compared with controls. After 4–5 years post-reconstruction, adduction moments are higher than for uninjured controls. Peak knee flexion moments are still lower after 5 years of surgery in ACLR participants than in uninjured controls.

The differences for knee moments indicate that the knee function is not fully restored following reconstruction, and long-term maintenance programs may be needed.

1 Introduction

Ruptures of the anterior cruciate ligament (ACL) of the knee are considered to be one of the most serious sport- and exercise-related injuries in terms of the patients' experiences [1] and health costs [2]. The injury is most common in sports such as netball, football, basketball, and handball, and is often associated with non-contact mechanisms such as jump landing and pivoting or twisting movements [3]. ACL injuries are treated surgically with reconstructions plus rehabilitation or non-surgically with rehabilitation alone [4]. The overall incidence of ACL reconstruction (ACLR) has been reported as 23 % within 3 years of injury [5]. A recent systematic review found that, following ACLR, 81 % of patients return to some form of sport, but only 55 % return to competitive sport [6]. In the long term, 50–70 % of patients with ACL rupture develop symptoms associated with post-traumatic osteoarthritis (PTOA) after 10 years, adding further towards the personal and health costs of these injuries [7]. Patients who undergo ACLR appear to have a slightly higher risk of developing symptoms associated with knee osteoarthritis (OA) than those managed non-surgically [8]. However, this evidence is limited, and the prevalence of OA between individuals treated with surgical reconstructions compared with non-surgical management is still unclear [8–10].

Altered gait biomechanics, as indicated by joint angles and moments, may indicate unresolved impairments following injury and surgery, and may also be a mechanical contributing factor towards the development of PTOA [11, 12]. Differences for these biomechanical variables appear to be important when comparing the injured side with the contralateral uninjured side in patients with ACLR and also with control groups. Individuals with ACLR have been shown to walk with greater peak knee flexion angles than controls in the early phases (<6 months), but with less flexion (more extended) after 1 year post-surgery [13]. In terms of joint loading in the frontal plane, external knee adduction moments are often used as the measure of load distribution in the frontal plane [14]. Increased peak adduction moments during walking have been associated with progression [12, 15] and severity of medial tibio-femoral compartment OA [16]. Studies have reported conflicting results for peak adduction moments in participants with ACLR, varying from low to high magnitude [17, 18]. In the sagittal plane, lower external flexion moments were found following ACLR at mid-stance of walking at 2–12 months post-operatively compared with controls [19], while no significant differences were reported on average 17 months post-surgery [20]. Decreased flexion moments while walking and during stair navigation have also been found in patients with knee OA. Furthermore, a recent systematic review found increased flexion moments during walking in patients with knee OA following joint replacements, compared with pre-operative measures [21]. Flexion moments require an internal quadriceps moment to counteract the forces of body weight [22]. Decreased flexion moments are likely to indicate unloading of the knee joint or a compensatory strategy, possibly due to decreased neuromuscular control, lingering symptoms, and hesitancy for weight-bearing on that side [23].

While several studies have explored gait biomechanics following ACLR, various methodologies and tasks have been used. Summarizing these gait biomechanics will increase the understanding of possible impairments and guide rehabilitation interventions to improve outcome and decrease long-term disability. Previous reviews on recovery of gait biomechanics following ACLR have either not included meta-analysis [24] or analyzed only sagittal plane moments [25]. Hart et al. [13] recently reviewed gait analysis only during walking activity, finding deficits in sagittal plane moments but not in the frontal plane. The current review includes other activities of daily living, namely stair ascent, descent, and running, in addition to walking. It is possible that long-term impairments and asymmetries following ACLR may be more evident during those cyclical and repetitive activities that are associated with higher joint loading than walking [26]. Thus, the primary aim of this systematic review and meta-analysis

was to compare knee kinematics and kinetics in individuals with ACLR with the contralateral limb and with healthy age-matched controls during three different tasks (walking, stair ascent/descent, and running). The secondary aim was to describe the progression over time of those biomechanical variables following ACLR.

2 Methods

2.1 Search Strategy and Study Selection

Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were followed as appropriate for observational studies [27]. Embase, MEDLINE, Web of Science, and Scopus databases were searched from their original available dates to 1 July 2014, with an update on 10 August 2015. Keywords used and the search strategy are presented in Table 1.

The following inclusion criteria were applied: (1) observational study designs (prospective cohort studies, case-control studies, and cross-sectional studies) or randomized clinical trials (RCTs) with the comparison with control group or comparison with the contralateral uninjured side at baseline; (2) men and women, with isolated ACL ruptures or in addition to menisci and other ligamentous injury of the knee, managed with reconstructions with any type of graft; (3) studies estimating kinematic and kinetic data (discrete variables: peak angles in the three planes of movement, peak external knee flexion, and adduction moments) of the ACLR knee and the asymptomatic controls during walking, stair negotiation, or running. Studies focusing on other lower limb injuries, or

other tasks, such as jumping or pivoting, were excluded. No language restrictions were applied. Two reviewers (MK and GS) screened all relevant titles and abstracts and were blinded to authors and journals. Full-text articles were screened according to the pre-set inclusion and exclusion criteria by the two reviewers. Any discrepancies were resolved through consensus, and a third reviewer was available if needed (DR). A manual search of the reference lists of the included articles was also conducted.

2.2 Risk of Bias Within Included Studies

A modified version of the Downs and Black Quality Index [28] was used to assess the risk of bias within included studies. The modified checklist included 16 questions from the following sub-groups: reporting (items 1, 2, 3, 5, 6, 7, and 10), external validity (items 11 and 12), internal validity bias (item 15, 16, 18, and 20), internal validity confounding (selection bias) (items 21 and 25), and power (item 27) (Electronic Supplementary Material [ESM] Table S1). Item 25 relates to confounding factors: walking or running speed were considered as the most relevant confounders, as these variables have been shown to be correlated with external knee adduction and flexion moments [29]. Thus, a score of “1” was applied if speed of ambulation was not significantly different between participant groups, or if appropriate statistical analyses were performed, including speed as a covariate. Item 27 was reworded as “Were appropriate power calculations reported?” and the score was changed from 0–5 to a scale of 0 or 1. The study was assigned a score of “1” if prior power analysis was performed, with no scores being assigned in the absence of power analysis. The risk of bias assessment was performed independently by two reviewers (MK and GS), with discrepancies resolved during consensus meetings. Kappa’s correlation coefficient was calculated to determine the agreement of the scores of the individual questions for each study between the two reviewers. The modified Downs and Black Quality Index was scored out of 17. Studies with scores of 11 and above (65 %) were considered to have a low risk of bias, and studies with scores below 11 were considered to have a high risk of bias [30].

2.3 Data Extraction and Meta-Analysis

Data extracted from each study included sample size, participants’ demographics, type of functional activities, and discrete variables for knee kinematics and kinetics (i.e., peak angles and moments during the stance phase of gait). For external adduction moments, the first of the two peaks was extracted. Peak knee tibial rotations were reported during the stance phase of gait. We contacted the corresponding authors when relevant data were not presented or were published as graphs.

Table 1 Keywords used for the data search

Anterior cruciate ligament	Reconstruction	Gait	Kinetics
ACL	Injury	Walking	Kinematics
Knee joint	Tear	Jogging	
	Graft	Running	
	Surgery	Locomotion	
	Deficiency	Ambulation	
		Stair climbing	
		Stair ascent	
		Stair descent	
		Stair negotiation	
		Gait analysis	

The search terms within each group were combined with the ‘OR’ operator; the ‘AND’ operator was used to combine the results from all four groups to obtain final yield

ACL anterior cruciate ligament

We used Review Manager v5.3 (RevMan) (Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2014) to perform the meta-analysis when more than one study investigated the same dependent variable during a given task using three-dimensional movement analysis. The main aim of the forest plot was to determine (standardized) mean differences and confidence intervals. Effect sizes (ESs) and 95 % confidence intervals (CIs) were calculated for between-group (ACLR groups vs. uninjured control groups) and within-group (injured sides vs. contralateral uninjured sides of the ACLR groups) comparisons of kinematic data and kinetic data. Calculated individual or pooled ESs were categorised as small (<0.5), medium (≥ 0.5 and <0.8), or large (≥ 0.8) [31]. The level of statistical heterogeneity for pooled data was established using γ^2 and I^2 statistics (heterogeneity defined as $p < 0.05$). For peak joint angles, we also calculated mean differences (MDs) (net difference in peak angle among any two groups) in degrees for findings of individual studies or pooled data from two or more studies, as these may be more easily translated to clinical practice than ESs.

To conduct between- and within-group comparisons of the biomechanical variables over time, the included studies were ordered in the forest plots according to “time since surgery” (in weeks or months). The shortest time since surgery was presented at the top, and the longest average time since surgery was presented at the bottom of the forest plot. This approach allowed a qualitative analysis of differences in biomechanical variables between and within groups over time.

2.4 Level of Evidence

The level of evidence was defined for each variable as per Van Tulder et al. [32] as follows. (1) Strong evidence provided by pooled results derived from three or more studies, including a minimum of two studies with a low risk of bias, that are homogenous ($p > 0.05$); the analysis may be associated with a statistically significant or non-significant pooled results. (2) Moderate evidence provided by statistically significant pooled results derived from multiple studies, including at least one study with low risk of bias, that are statistically heterogeneous ($p < 0.05$), or from multiple studies with a high risk of bias that are statistically homogenous ($p > 0.05$). (3) Limited evidence provided by results from multiple studies with a high risk of bias that are statistically heterogeneous ($p < 0.05$), or from one study with a low risk of bias. (4) Very limited evidence provided by results from one study with a high risk of bias. (5) Conflicting evidence provided by pooled results that are insignificant and derived from multiple studies regardless of quality that are statistically heterogeneous ($p < 0.05$, i.e., inconsistent).

3 Results

3.1 Search Results

The first search yielded 1550 articles (Fig. 1); 65 remained after excluding duplicates and irrelevant titles and screening abstracts. A further 25 studies were excluded based on the inclusion and exclusion criteria, giving a final yield of 40 studies that were included in the qualitative analysis. Of the studies, 37 were in English, two in German [33, 34], and one in Chinese [35]. Data from 27 studies could be included in meta-analyses.

3.2 Risk of Bias

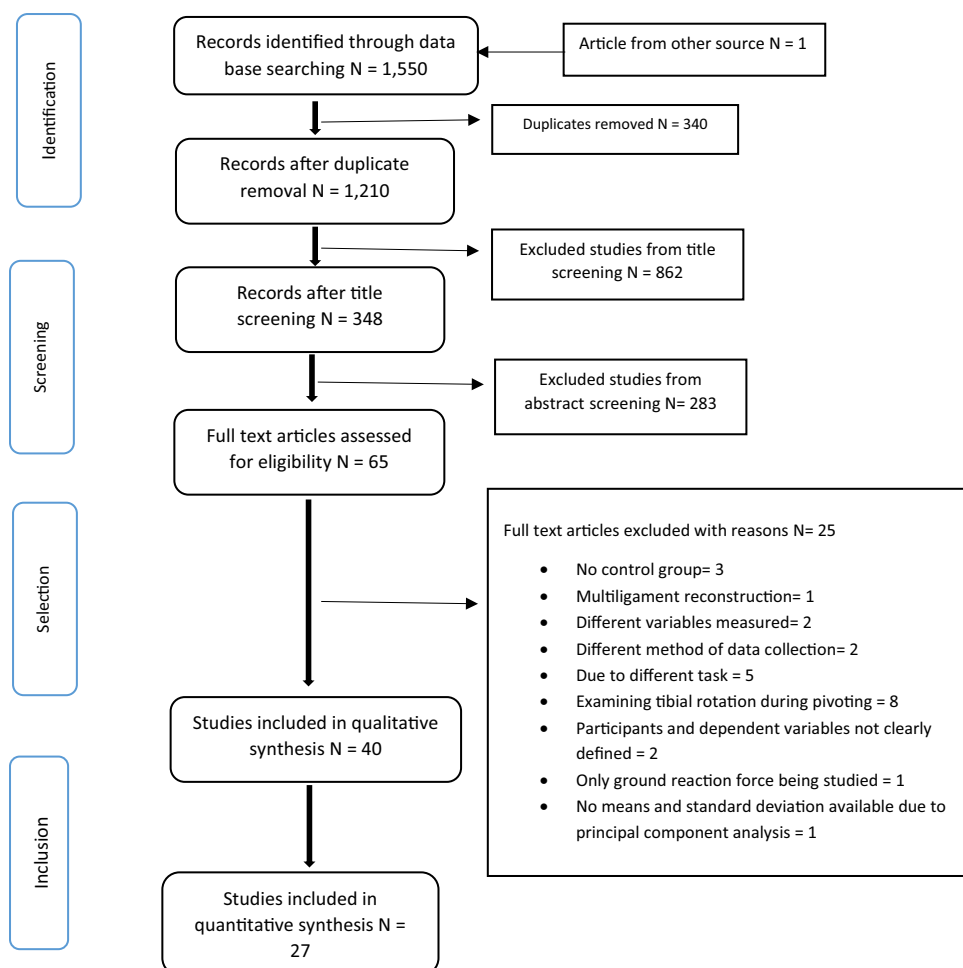
Kappa's correlation coefficient was 0.93 for the agreement between the two reviewers for risk of bias assessment. Of the 40 studies, 26 were rated as low risk and 14 as high risk (Table 2). Seven corresponding authors were contacted to request data for dependent variables [36–42]; one provided the details [38]. Variables were calculated from the figures when corresponding authors did not respond [43–45].

3.3 Overview of Included Studies

A total of 31 studies included comparisons with asymptomatic age-matched subjects, and nine with the contralateral knee of this ACLR participant group (Table 3). Kinematic and kinetic variables were studied during walking in 37 studies at self-selected comfortable speed, during stair ascent and descent in eight studies [42, 43, 46–51], and during running in five studies [41, 43, 52–54]. The number of included participants ranged from eight with ACLR and ten controls [41, 52] to 45 in both groups [51], and time since surgery ranged from 2 weeks [48] to 18 years [47].

An RCT conducted by Hooper et al. [48] was included and, for the purpose of this review, the baseline comparisons for the injured versus the contralateral uninjured knees were considered for the meta-analysis. A longitudinal cross-sectional study by Webster et al. [55] was included, with the results being reported at 1 year and 3.3 years. Data from both of these time periods were included for meta-analysis.

Grand means and standard deviations (SDs) were calculated [56] where studies presented results for different groups with ACLR. In one study [48], baseline results of participants undergoing either closed or open kinetic training were combined. Where studies compared outcomes of different surgical procedures, specifically, between hamstring and patella tendon grafts [57, 58], single- and double-bundle grafts [59], and use of different

Fig. 1 Flow diagram for study-selection process

portal techniques [60], the results of the ACLR groups were combined and compared with those of the controls. Lewek et al. [52] included two groups of participants based on quadriceps strength: (1) “strong ACLR” group and (2) “weak ACLR” group. The data of these two groups were combined into a grand mean and SD. Furthermore, variables for participants with ACLR for the dominant and non-dominant limbs were combined and compared with the control group in a study by Wang et al. [39]. Results of two groups reported by Di Stasi et al. [61], participants who passed ‘return to sports’ criteria and those who did not pass, were also combined into grand means and SDs, calculated from the published 95 % CIs. In another study by Di Stasi et al. [45], means and CIs were estimated from the given figure and grand mean and SDs for injured and uninjured sides were calculated, combining the results for men and women. Roewer et al. [44] conducted a longitudinal study, reporting data at 6 months and at 2 years post-reconstruction. CIs were estimated from the figure, and SDs were calculated. Data reported at both time periods were included in the forest plots [44].

3.4 Meta-Analyses

Meta-analyses were conducted for peak angles and moments (peak knee flexion and adduction moments) during the stance phase of gait (Figs. 2, 3, 4, 5, 6, 7). Tables 4 and 5 provide the ESs and CIs for the meta-analyses and for variables that were explored in one study only. Mean differences for joint angles are provided in Table S2 in the ESM.

3.4.1 Joint Angles

Peak knee flexion angles during walking were reported in 11 studies [44, 47, 48, 50, 52, 54, 55, 57, 61–63]. The meta-analysis indicated strong evidence for no difference compared with controls [47, 52, 54, 57, 62, 63] and moderate evidence for less flexion for the ACLR compared with the contralateral limb (ES -0.06 ; 95 % CI -0.8 to -0.4 ; MD 4.3°) [44, 47, 48, 55, 61]. A sensitivity test was performed by excluding a study by Hooper et al. [48], as this study compared ACLR participants at 2 weeks post-

Table 2 Methodological quality assessment

Study	Categories and questions																Power	Total	%	Risk of bias
	Reporting								External validity		Internal validity									
Original numbering	1	2	3	5	6	7	10	11	12	15	16	18	20	21	25	27				
New numbering	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
Bulgheroni et al. [20]	1	1	1	0	1	0	0	U	U	U	1	0	1	1	1	0	8	47	HR	
Butler et al. [40]	1	1	1	2	1	1	0	0	0	U	1	1	1	1	1	1	13	76	LR	
Bush-Joseph et al. [43]	1	1	1	1	1	1	1	0	0	U	1	1	0	1	1	0	11	65	LR	
Button et al. [86]	1	1	1	2	1	1	1	0	0	U	1	1	0	1	1	0	12	71	LR	
Czamara et al. [59]	1	1	1	2	1	1	1	0	0	0	1	1	1	1	0	0	12	71	LR	
DeVita et al. [91]	1	1	1	1	1	1	0	0	0	U	1	1	0	1	0	0	9	53	HR	
Di Stasi et al. [61]	1	1	1	1	1	1	1	0	0	U	1	1	1	NA	1	0	11	69	LR	
Di Stasi et al. [45]	1	1	1	1	1	1	1	0	0	0	1	1	1	NA	1	0	11	69	LR	
Ferber et al. [62]	1	1	1	1	1	1	0	0	0	U	1	1	0	0	0	0	9	53	HR	
Ferber et al. [92]	1	1	1	1	1	1	0	0	0	U	1	1	0	1	1	1	11	65	LR	
Gao and Zheng [37]	1	1	1	1	1	1	0	U	U	U	1	1	1	1	1	0	11	65	LR	
Gao et al. [50]	1	1	1	2	1	1	0	0	0	1	1	1	1	1	0	0	12	71	LR	
Georgoulis et al. [63]	1	1	1	1	1	1	0	0	0	U	1	1	1	1	1	0	11	65	LR	
Hall et al. [47]	1	1	1	2	1	1	1	0	0	U	1	1	1	1	1	0	13	76	LR	
Hooper et al. [48]	1	1	1	1	1	1	0	0	0	U	1	1	0	NA	0	0	8	50	HR	
Hooper et al. [49]	1	1	1	1	1	1	1	0	0	U	1	1	0	NA	0	0	9	56	HR	
Karimi et al. [65]	1	1	1	1	1	1	1	U	U	U	1	1	0	1	1	0	11	65	LR	
Knoll et al. [82]	1	1	1	1	1	1	0	0	0	U	1	1	0	0	1	0	9	53	HR	
Kowalk et al. [46]	1	1	1	1	1	1	1	0	0	U	1	1	1	1	0	0	11	65	LR	
Kuenze et al. [41]	1	1	1	2	1	1	1	0	0	0	1	1	1	1	1	1	14	82	LR	
Lewek et al. [52]	1	1	1	1	1	1	1	0	0	U	1	1	0	0	1	0	10	59	HR	
Noehren et al. [54]	1	1	1	2	1	1	1	0	0	U	1	1	0	1	1	0	12	71	LR	
Patterson et al. [18]	1	1	1	2	1	1	1	0	0	U	1	1	1	1	0	0	12	71	LR	
Roewer et al. [44]	1	1	1	1	1	1	1	0	0	U	1	1	1	NA	1	0	10	69	LR	
Sato et al. [53]	1	1	1	2	1	1	0	0	0	1	1	1	1	1	0	0	12	69	LR	
Scanlan et al. [36]	1	1	1	1	1	1	0	0	0	0	1	1	1	1	0	0	10	56	HR	
Schmalz et al. [33]	1	1	1	1	0	0	0	0	0	0	1	1	0	1	0	0	7	41	HR	
Schmalz et al. [34]	1	1	1	1	0	0	0	0	0	0	1	1	0	1	1	0	8	47	HR	
Schroeder et al. [42]	1	1	1	1	1	1	1	0	0	0	1	1	0	1	0	0	10	59	HR	
Tellini et al. [93]	1	1	1	2	1	1	1	0	0	U	1	1	0	1	0	0	10	59	HR	
Timoney et al. [19]	1	1	1	1	1	1	0	0	0	U	1	1	0	1	1	0	10	59	HR	
Varma et al. [38]	1	1	1	2	1	1	1	0	0	U	1	1	0	1	1	0	12	71	LR	
Wang et al. [35]	1	1	1	1	1	0	0	0	0	U	1	1	0	1	1	0	10	63	HR	
Wang et al. [39]	1	1	1	1	1	1	1	0	0	U	1	1	1	1	0	0	11	65	LR	
Wang et al. [60]	1	1	1	1	1	1	0	0	0	U	1	1	1	1	1	1	12	71	LR	
Webster and Feller [57]	1	1	1	2	1	1	1	0	0	U	1	1	1	1	1	0	13	76	LR	
Webster and Feller [58]	1	1	1	2	1	1	1	U	U	U	1	1	1	1	1	0	13	76	LR	
Webster et al. [55]	1	1	1	1	1	1	1	0	0	U	1	1	1	1	0	0	11	65	LR	
Webster and Feller [17]	1	1	1	2	1	1	0	0	0	U	1	1	1	1	1	0	12	71	LR	
Zabala et al. [51]	1	1	1	1	1	1	1	0	0	U	1	1	1	1	0	0	11	65	LR	

HR high risk of bias, LR low risk of bias, NA not applicable, U unable to determine, 1 clear aim/hypothesis, 2 outcome measures clearly described, 3 patient characteristics clearly described, 4 confounding variables described, 5 main findings clearly described, 6 measures of random variability provided, 7 actual probability values reported, 8 participants asked to participate representative of entire population, 9 participants prepared to participate representative of entire population, 10 blinding of outcome measurer, 11 analysis completed was planned, 12 appropriate statistics, 13 valid and reliable outcome measures, 14 appropriate case control matching, 15 adjustment made for confounding variables, 16 power

Table 3 Study characteristics and variables measures at knee in included studies

Study	ACL group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Bulgheroni et al. [20]	Participants: 15 (M) Age: 25 (3) Time between injury and surgery: NR Surgical procedure: BPTB Time since surgery: 17 (5) mo	Participants: 5 (M) Age: 28 (3)	4 Cameras, 100 Hz Kistler force plate, 500 Hz	Walking: natural cadence (112 ± 5.1 steps/min); footwear condition NR. Average linear envelopes only. Kinematics: flexion/extension; adduction/abduction; internal/external rotation Kinetics: external moments; flexion/extension; adduction/abduction; internal/external rotation Ground reaction forces: Fz, Fx, Fy
Butler et al. [40]	Participants: 17 (4 M, 13 F) Age: 23.6 (5.8) Weight: 24.0 (3.2) Time between injury and surgery: NR Surgical procedure: NR Time since surgery: 5.3 (4.4) yr	Participants: No. NR Age: 23.4 (5.7) Weight: 23.3 (2.4)	8-Camera motion analysis system, 80 Hz Force plate, 1050 Hz	Walking: “Intentional” walking speed, monitored by average velocity of L5/S1 marker; neutral laboratory shoes Kinematics: peak adduction during stance; total adduction/abduction excursion (range) Kinetics: internal moments peak abduction moment
Bush-Joseph et al. [43]	Participants: 22 (13 M, 9 F) Age: 27 (11) Same weight in both groups Time between injury and surgery: 8 months Surgical procedure: BPTB Time since surgery: 22 (12) mo	Participants: 22 (13 M, 9 F) Age: 29 (8) Weight: same in both groups	Multi-camera optoelectronic Marker Force plate	Walking: speed 1.18 (0.018) m/s in ACLR and 1.16 (0.016) m/s in control group; foot wear condition NR Kinetics: external moment; flexion and extension Stair climbing Kinetics: peak external flexion moment Running speed 2.70 (0.39) m/s in ACLR and 2.58 (0.042) m/s in control group Kinetics: external flexion moment
Button et al. [86]	Participants: 21 (16 M, 5 F) Age: 29.1 (9) Weight: 80.1 (9.5) Time since surgery: 13.5 (9) mo Surgical procedure: single bundle gracilis-ST graft	Participants: 21 (12 M, 5 F) Age: 26.8 (7.7) Weight: 77.6 (19.6)	8-Camera Vicon motion analysis system, 250 Hz Kistler force plate, 1000 Hz	Walking: “Normal” walking speed; footwear condition NR Kinematics: knee flexion/extension Kinetics: peak internal extension moment; peak external adduction moment Gait velocity; knee “fluency”
Czamara et al. [59]	SB ACLR ^b Participants: 10 (M) Age: 26 (10.22) Weight: 82.20 (9.19) Time between injury and surgery: 4 (2) mo Time since surgery: 13.60 (3.27) mo Surgical procedure: single bundle HT graft DB ACLR ^b Participants: 13 (M) Age: 28 (8.14) Weight: 76.31 (11.26) Time between injury and surgery: 4 (2) mo Time since surgery: 14.08 (5.42) mo Surgical procedure: double bundle HT graft	Participants: 15 M Age: 23.40 (2.41) Weight: 76.27 (7.09)	6-Infrared camcorders, BTS SMART system, 120 Hz 2 Kistler force plates, 960 Hz	Walking: “Normal” walking speed; footwear condition NR Kinematics: range of rotation during different stance and swing phase of gait cycle

Table 3 continued

Study	ACLR group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Devita et al. [91]	Participants: 8 (5 M, 3 F) Age: 20.3 Weight: 74.2 Time between injury and surgery: NR Time since surgery: 3 weeks, 6 months Surgical procedure: BPTB	Participants: 22 (14 M, 8 F) Age: 21.7 Weight: 73.7	One video camera, 60 Hz Force plate, 1000 Hz	Walking: instructions and footwear condition NR Kinematics: ROM in swing; average position in stance; extensor angular impulse Kinetics: extension/flexion joint torque; extension/flexion joint power
Di Stasi et al. [61]	Participants: 42 (30 M, 12 F) Age: 29.3 (10.8) Time between injury and surgery: 17.2 weeks Time since surgery: 6 months Surgical procedure: HT/gracilis allograft	Contralateral limb as controls	8-Camera system, 120 Hz Force plate, 1080 Hz	Walking: self-selected comfortable speed controlled via infrared photocells; footwear condition NR Kinematics: flexion/extension angles, from initial contact to peak flexion Kinetics: internal joint moments and joint power
Di Stasi et al. [45]	Participants: 39 (27 M, 12 F) Age: M 28 (10), F 32 (12) Time between injury and surgery: NR Time since surgery: 6 months Surgical procedure: HT allograft	Contralateral limb as controls	8-Camera system, 120 Hz Force plate, 1080 Hz	Walking: self-selected comfortable speed controlled via infrared photocells; footwear condition NR Kinematics: joint excursions during mid-stance Kinetics: internal joint extension moment
Ferber et al. [62]	Participants: 10 (5 M, 5 F) Age: 27.7 (9.1) Weight: 79.1 (13.8) Time between injury and surgery: NR Time since surgery: 3 months Surgical procedure: BPTB	Participants: 10 (5 M, 5 F) Age: 24.4 (3.1) Weight: 67.2 (10.7)	4 Cameras, 120 Hz Force plate	Walking: self-selected comfortable speed controlled via metronome; footwear condition NR Kinematics: flexion/extension angles Kinetics: flexor/extensor joint moments; power Ground reaction forces: Fz, Fx, Fy
Ferber et al. [92]	Participants: 10 (5 M, 5 F) Age: 27.7 (9.1) Weight: 79.1 (13.8) Time between injury and surgery: NR Time since surgery: 3 months Surgical procedure: BPTB	Participants: 10 (5 M, 5 F) Age: 24.4 (3.1) Weight: 67.2 (10.7)	4 Cameras, 120 Hz Force plate, 1200 Hz	Walking: self-selected comfortable speed controlled via metronome; footwear condition NR Kinematics: average flexion/extension angles Kinetics: extensor moments/ extensor angular impulse; work and power
Gao and Zheng [37]	Participants: 14 (12 M, 2 F) Age: 25.1 (5.9) Weight: 82.5 (15.0) Time between injury and surgery: NR Time since surgery: 3 (12) mo Surgical procedure: 7 HT, 5 BPTB, 2 Achilles tendon	Participants: 15 (12 M, 3 F) Age: 22.8 (2.6) Weight: 76.8 (16.4)	11-Camera stereo photogrammetric system, 60 Hz 2 force plate	Walking: self-selected comfortable speed; barefooted Kinematics: flexion, tibial internal rotation, varus angles during static standing posture; mean curves for knee flexion/extension; varus/valgus, axial rotation, and for translation in the three planes Spatiotemporal variables: step and stride length; step and stride speed; 1st and 2nd single and double-support phase

Table 3 continued

Study	ACL group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Gao et al. [50]	Participants: 12 (10 M, 2 F) Age: 24.8 (6.1) Weight: 83.2 (16.0) Time between injury and surgery: NR Time since surgery: 3 (12) mo Surgical procedure: 6 HT, 4 BPTB, 2 Achilles tendon	Participants: 12 (10 M, 2 F) Age: 23.3 (2.5) Weight: 78.8 (15.8)	11 Eagle camera motion capture system, 60 Hz Force plate	Stair ascent/descent: self-selected comfortable speed; barefooted Kinematics: flexion, internal rotation, varus; peak and total flexion/extension; varus/valgus IE rotation peak for ascent and for descent; Mean curves for flexion/extension, varus/valgus, axial rotation for ascent and for descent Spatiotemporal variables: stance/swing time; total time; 1st and 2nd single and double support; CTO/CFS/ TO Walking: self-selected pace; footwear condition NR Kinematics: flexion at TO and HS; max flexion during swing and loading response; max tibial abduction/adduction; max tibial IE rotation Spatiotemporal: cadence; average gait velocity
Georgoulis et al. [63]	Participants: 21 (19 M, 2 F) Age: 25 (4) Weight: 69.11 (7.89) Time between injury and surgery: NR Time since surgery: 30 (16.9) wk Surgical procedure: BPTB	Participants: 10 (8 M, 2 F) Age: 24.7 (3.7) Weight: 62.1 (12.38)	6 Cameras, 50 Hz	Walking: self-selected pace, with preferred shoes Kinematics: initial flexion angle Kinetics: peak internal extension moment; peak external varus moment Stair ascent/descent: Kinematics: initial flexion angle Kinetics: peak extension moment; peak varus moment Walking: gait speed and footwear NR Kinematics: flexion at HS, TO, MS excursion Kinetics: concentric /eccentric energy; flexor/extensor impulse; flexion/extension moment Stair ascent: Kinematics: flexion at HS Kinetics: peak extensor moment; extensor impulse; peak concentric power; concentric energy Stair descent: Kinematics: flexion at HS Kinetics: peak extensor moment; extensor impulse; peak eccentric power; eccentric energy
Hall et al. [47]	Participants: 15 (7 M, 8 F) Age: 26 (6) Weight: 75 (15) Time between injury and surgery: NR Time since surgery: 6 (2–18) yr Surgical procedure: 41 % HT, 41 % patellar tendon, 12 % cadaver + hamstring, 6 % HT	Participants: 17 (7 M, 10 F) Age: 26 (4) Weight: 65 (12)	8-Camera 3D motion analysis system, 160 Hz Force plate, 1600 Hz	Walking: self-selected pace, with preferred shoes Kinematics: initial flexion angle Kinetics: peak internal extension moment; peak external varus moment Stair ascent/descent: Kinematics: initial flexion angle Kinetics: peak extension moment; peak varus moment Walking: gait speed and footwear NR Kinematics: flexion at HS, TO, MS excursion Kinetics: concentric /eccentric energy; flexor/extensor impulse; flexion/extension moment Stair ascent: Kinematics: flexion at HS Kinetics: peak extensor moment; extensor impulse; peak concentric power; concentric energy Stair descent: Kinematics: flexion at HS Kinetics: peak extensor moment; extensor impulse; peak eccentric power; eccentric energy
Hooper et al. [48]	CKC group^b Participants: 18 (13 M, 5 F) Age: NR Weight: 75.1 (12.3) Time between injury and surgery: 50.3 (61.8) mo Time since surgery: 2 weeks Surgical procedure: BPTB OKC group^b Participants: 19 (16 M, 3 F) Age: NR Weight: 77.4 (15.3) Time between injury and surgery: 34.1 (30.4) mo Time since surgery: 2 weeks Surgical procedure: BPTB	Contralateral limb as controls	3 Cameras, 50 Hz Force plate, 200 Hz	Walking: self-selected pace, with preferred shoes Kinematics: initial flexion angle Kinetics: peak internal extension moment; peak external varus moment Stair ascent/descent: Kinematics: initial flexion angle Kinetics: peak extension moment; peak varus moment Walking: gait speed and footwear NR Kinematics: flexion at HS, TO, MS excursion Kinetics: concentric /eccentric energy; flexor/extensor impulse; flexion/extension moment Stair ascent: Kinematics: flexion at HS Kinetics: peak extensor moment; extensor impulse; peak concentric power; concentric energy Stair descent: Kinematics: flexion at HS Kinetics: peak extensor moment; extensor impulse; peak eccentric power; eccentric energy

Table 3 continued

Study	ACLR group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Hooper et al. [49]	ACL-6 group Participants: 8 (5 M, 3 F) Age: 33.4 (11.0) Weight: 77.4 (14.7) Time between injury and surgery: NR Time since surgery: 6.4 (0.5) mo Surgical procedure: 6 BTB, 2 BTB with LAD ACL-12 group^c Participants: 9 (4 M, 5 F) Age: 34.2 (10.4) Weight: 71.1 (15.7) Time between injury and surgery: NR Time since surgery: 11.9 (0.6) mo Surgical procedure: 5 BTB, BTB with LAD	Contralateral limb as controls	3 Infrared cameras, 50 Hz Force plate, 200 Hz	Walking: gait speed and footwear condition NR Kinematics: flexion at HS, TO, MS excursion Kinetics: peak external flexion/extension/varus moment; peak concentric power during MS Stair ascent: Kinematics: flexion at toe contact and TO Kinetics: peak external flexion/varus moment; peak concentric power Stair descent: Kinematics: flexion at toe contact and TO Kinetics: peak external flexion/varus moment; peak eccentric power
Karimi et al. [65]	Participants: 15 (sex NR) Age: 33 (2.6) Time between injury and surgery: NR Time since surgery: 6 months Surgical procedure: combined HT and peroneus longus tendon	Participants: 15 (sex NR) Age: 32 (3), weight and sex matched control group	7 Cameras Force plate	Walking: self-selected normal walking speed; foot wear NR Kinematics: hip, knee, ankle and pelvis range of motion in three planes Kinetics: knee flexion/ extension moment, adduction moment, internal and external rotation moment Spatiotemporal gait: Stride length, speed and cadence Ground reaction force
Knoll et al. [82]	Participants: 16 (9 M, 7 F) Age: M 39.71 (2.10), F 30.31 (10.48) Weight: M 88.10 (20.2), F 62.11 (8.38) Time between injury and surgery: 12 days and 28.2 months Time since surgery: 12 months Surgical procedure: BPTB	Participants: 51 (31 M, 20 F) Age: M 28.17 (7.69), F 25.09 (4.21) Weight: M 77.89 (11.88), F 59.86 (6.38)	Ultrasound based zebris, 100 Hz Force plate	Walking (on treadmill): 0.83 m/s; barefoot Kinematics: peak flexion/extension Spatiotemporal: step length; walking base
Kowalk et al. [46]	Participants: 7 (5 M, 2 F) Age: 18–38 Time between injury and surgery: NR Time since surgery: 6 (3.2–11.3) mo Surgical procedure: BPTB	Participants: 10 (6 M, 4 F) Age: 20–49	4 Cameras, 60 Hz Kistler force plate, 60 Hz	Stair ascent: freely selected speed, barefoot Kinematics: maximum joint excursion Kinetics: internal moments, power; work

Table 3 continued

Study	ACL group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Kuenze et al. [41]	Participants: 20 (11 M, 9 F) Age: 22.7 (5.2) Weight: 72.7 (13.7) Time between injury and surgery: NR Time since surgery: 33.9 (23.4) mo Surgical procedure: 11HT, 9 BPTB Strong ACLR ^b Participants: 8 (5 M, 3 F) Age: 21.4 (6.0) Time between injury and surgery: NR Time since surgery: 14.3 (2.4) wk Surgical procedure: ST-gracilis graft Weak ACLR ^b Participants: 10 (4 M, 6 F) Age: 25 (7.8) Time between injury and surgery: NR Time since surgery: 20.8 (9.4) weeks Surgical procedure: ST-gracilis graft	Participants: 23 (12 M, 11 F) Age: 21.9 (3.6) Weight: 69.6 (13.8) Participants: 10 (8 M, 2 F) Age: 32.2 (6.6)	12 Cameras, 250 Hz Force plate imbedded in treadmill, 1000 Hz 6 Cameras, 120 Hz Force plate, 960 Hz Isokinetic dynamometer	Running: 2.68 m/s, footwear condition NR Kinematics: external hip, knee and ankle kinematics during stance and swing phase of the gait in sagittal and frontal plane Kinetics: external hip, knee and ankle flexion moments during stance and swing phase of the gait in sagittal and frontal plane Walking: free speed, velocity monitored with photoelectric beam; footwear condition NR Kinematics: initial contact; peak flexion Kinetics: peak internal flexion moments Running: free speed, velocity monitored with photoelectric beam Kinematics: initial contact; peak flexion Kinetics: peak internal flexion moments
Noehren et al. [54]	Participants: 20 F Age: 25 (6.2) Weight: 64 (8.2) Time between injury and surgery: NR Time since injury: 5.2 (3.2) yr Surgical procedure: 8 HT 7 BPTB, 5 autograft	Participants: 20 F Age: 26 (5.1) Weight: 61 (5.1)	15-Camera motion analysis system, 200 Hz Force plate, 1200 Hz	Walking: 1.5 m/s Kinematics: flexion Kinetics: impact force, Average loading rate, extensor moment, ground reaction forces: Fz, Fx, Fy Run: 2.9 m/s Kinematics: flexion Kinetics: impact force Average loading rate, extensor moment, ground reaction forces: Fz, Fx, Fy Walking: self-selected "normal" walking speed; barefoot Kinematics: transverse, sagittal and frontal angles throughout entire stride Kinetics: external peak adduction moment
Patterson et al. [18]	Participants: 17 (3 M, 14 F) Age: 23.7 (3.12) Weight: 64.9 (9.02) Time between injury and surgery: NR Time since injury: 3.50 (3.25) yr Surgical procedure: 8 HT, 9 BPTB	Participants: 17 F Age: 20.8 (1.17) Weight: 64.7 (7.06)	3 Cameras, 200 Hz 2 force plate, 1000 Hz CODA motion analysis system	Walking: self-selected "normal" walking speed; barefoot Kinematics: transverse, sagittal and frontal angles throughout entire stride Kinetics: external peak adduction moment
Roewer et al. [44]	Participants: 26 (18 M, 8 F) Age: 29.6 (10.7) Time between injury and surgery: NR Time since surgery: 24 months Surgical procedure: ST-gracilis graft, allograft	Contralateral limb as controls	8 Cameras, 120 Hz Force plate, 1080 Hz	Walking: self-selected "normal" walking speed controlled via infrared photo cells Kinematics: peak flexion angle, excursion angle Kinetics: internal knee extensor moment, power

Table 3 continued

Study	ACLR group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Sato et al. [53]	Participants: 7 (3 M, 4 F) Age: 21.3 (4.9) Weight: 59.0 (4.7) Time between injury and surgery: NR Time since surgery: 14.3 (1.8) mo Surgical procedure: HT	Contralateral limb as controls	9 Cameras, 200 Hz Force plate KT-1000 arthrometer	Walking: gait speed NR; barefoot Kinematics: maximum internal rotation angle Running: gait speed NR; barefoot Kinematics: maximum internal rotation angle
Scalan et al. [36]	Participants: 26 (11 M, 15 F) Age: 31 Weight: 68 Time between injury and surgery: NR Time since surgery: 24 months (7–65) mo Surgical procedure: 12 allograft (9 Achilles, 1 BPTB, 2 soft tissue allograft), 12 autograft (10 BPTB, 4 HT)	Contralateral limb as control	Optoelectronic motion capture system, 120 Hz Force plate	Walking: self-selected normal walking speed; foot wear NR Kinematics: IE rotation; VV rotation; flexion
Schmalz et al. [33]	Participants: 26 (sex NR) Age: 29 (6) Weight: 76 (10) Time between injury and surgery: NR Time since surgery: 8–52 weeks Surgical procedure: BPTB	Participants: 30 (sex NR) Age: 28 (5) Weight: 71 (9)	Optoelectronic motion capture system, 4 cameras, 100 Hz 2 Kistler force plates, 400 Hz	Walking: fast pace, footwear condition NR Kinematics: flexion/extension Kinetics: flexion/extension
Schmalz et al. [34]	Participants: 35 (sex NR) Age: 27 (7) Weight: 76 (12) Time between injury and surgery: NR Time since surgery: 8–52 weeks Surgical procedure: BPTB	Participants: 30 (sex NR) Age: 28 (5) Weight: 71 (9)	Optoelectronic motion capture system, 4 cameras, 100 Hz 2 Kistler force plates, 400 Hz	Walking: fast pace, footwear condition NR Kinematics: flexion/extension Kinetics: flexion/extension

Table 3 continued

Study	ACL group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Schroeder et al. [42]	ACL (HT) group^b Participants: 10 (sex NR) Age: 32.6 (8) Weight: 74.6 (7.4) Time between injury and surgery: NR Time since surgery: 9.4 (3.7) mo Surgical procedure: HT ACL (PT) group^b Participants: 6 Age: 30.8 (5.5) Weight: 74.8 (18) Time between injury and surgery: NR Time since surgery: 9.7 (4.3) mo Surgical procedure: PT	Participants: 9 age matched (sex NR) Age: 27.2 (3.7) Weight: 67.1 (13.4)	Passive motion capture system, 8 cameras, 120 Hz 3 force plates	Walking: speed and footwear condition NR Kinematics: flexion/extension, abduction/adduction and rotation Kinetics: peak flexion/extension, adduction/abduction and rotation moment Stair descent (step to floor): Kinematics: speed and footwear condition NR Kinetics: peak flexion/extension, adduction/abduction and rotation moment Stair descent (step to descent) Kinematics: speed and footwear condition NR Kinetics: peak flexion/extension, adduction/abduction and rotation moment
Tellini et al. [93]	Participants: 8 (5 M, 3 F) Age: 25.5 (7.1) Weight: 61.3 (27.1) Time between injury and surgery: NR Time since surgery: 32 (6) days Surgical procedure: BPTB	Participants: 8, age and sex matched	Video camera, 60 Hz	Walking: self-selected comfortable speed; footwear NR; walking on floor and on a foam mat Kinematics: knee angular displacement Spatiotemporal: stride length
Timoney et al. [19]	Participants: 10 (M) Age: 20–30 Time between injury and surgery: 30 (1–66) mo Time since surgery: 10 (9–12) mo Surgical procedure: BPTB	Participants: 10 Similar controls	Vicon motion analysis system, 200 Hz Force plate	Walking: free walking speed; barefoot Kinetics: ground reaction loading rate; maximally vertically directed loading rate; tibially directed loading rate; heel strike external knee moment; midstance external flexion moment Spatiotemporal: pre heel strike foot velocities; average walking velocity
Varma et al. [38]	Participants: 12 (9 M, 3 F) Age: 30.5 (8.68) Weight: 75 (11.13) Time between injury and surgery: NR Time since surgery: 4.5 (3.5) Surgical procedure: BPTB	Participants: 12 (9 M, 3 F) Age: 24.8 (8.81) Weight: 71.6 (11.2)	3-DGA, 10 cameras, 100 Hz Force plate, 1000 Hz	Walking: self-selected speed, barefoot Kinetics: peak adduction, flexion and extension moment Walking: uphill/downhill: Self-selected speed, barefoot Kinetics: peak adduction, flexion and extension moment
Wang et al. [35]	Participants: 29 (18 M, 11 F) Age: M 32.7 (4.6), F 30.2 (9.4) Weight: M 75.3 (9.1), F 55.0 (7.1) Time between injury and surgery: 2.6 months (2 weeks–16 months) Time since surgery: NR Surgical procedure: HT	Participants: 58 (36 M, 22 F) Age: 31.2 (5.2) Weight: 70.1 (24.2)	8-Camera motion analysis system, 60 Hz Measured before surgery, 3, 6, 9, 12 months after surgery	Walking: walking speed of 0.52 m/s; Footwear condition NR Kinematics: knee joint movement angle; knee joint angular acceleration; ACL relative motion parameters Spatiotemporal variables: step length; step width

Table 3 continued

Study	ACL R group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Wang et al., 2012 [39]	Group d^b Participants: 19 (12 M, 7 F) Age: 32.4 (8.6) Weight: 83.9 (18.8) Time between injury and surgery: NR Time since surgery: 14.1 (4.4) mo Surgical procedure: HT (16), BTB (3) Group n^b Participants: 22 (12, M, 10 F) Age: 31.1 (8.0) Weight: 81.4 (16.4) Time between injury and surgery: NR Time since surgery: 13.9 (5.5) mo Surgical procedure: HT (19), BTB (3)	Participants: 20 (13 M, 7 F) Age: 23.4 (3.0) Weight: 70.8 (13.2)	10 Cameras, 60 Hz Force plate	Walking: self-selected speed; footwear condition NR Kinematics: rotations/translations Kinetics: external adduction moment
Wang et al. [60]	TT^b Participants: 12 (7 M, 5 F) Age: 32.75 (7.2) Weight: 83.0 (16.7) Time between injury and surgery: NR Time since surgery: 17.1 (10.6) mo Surgical procedure: HT AMP^b Participants: 12 (7 M, 5 F) Age: 29.6 (7.2) Weight: 85.6 (16.5) Time between injury and surgery: NR Time since surgery: 8.8 (4.3) mo Surgical procedure: HT	Participants: 20 (15 M, 5 F) Age: 26.3 (7.7) Weight: 81.6 (15.5)	10 Cameras, 60 Hz 2 force plate, 1200 Hz	Walking: self-selected walking speed 1.10–1.37 m/s for control group, 0.92–1.45 m/s for ACLR group. Footwear condition NR Kinematics: at neutral stance: flexion/varus; femoral external rotation During walking: AP/ML/Superior-inferior ROM; peak valgus and external rotation Spatiotemporal: step and stride length; step and stride speed

Table 3 continued

Study	ACL group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Webster and Feller [57]	HT	Participants: 17 (16 M, 1 F) Age: 24.7 (5) Weight: 74.8 (12)	6-Camera Vicon motion analysis system, 50 Hz Force plate, 400 Hz	Walking: self-selected comfortable speed; barefoot. Kinematics: maximum flexion/extension; flexion at HS Kinetics: peak torque; max flexion/extension moment; extension moment at FS
	PT			
	Participants: 17 (16 M, 1 F) Age: 23.8 (5) Weight: 78.9(6) Time between injury and surgery: 10.6 (8.3) wk Time since surgery: 9.3 (2.2) mo Surgical procedure: HT			
	PT ^b			
	Participants: 18 (16 M, 2 F) Age: 26.6 (6) Weight: 79.3 (7.2) Time between injury and surgery: 11 (8.2) wk Time since surgery: 9.0 (2.4) mo Surgical procedure: HT	Participants: 18 (16 M, 2 F) Age: 24.7 (5) Weight: 74.5 (12.7)	6-Camera Vicon motion analysis system, 50 Hz Markers (Plug in gait) Force plate	Walking: self-selected comfortable speed; footwear condition NR Kinematics: varus/valgus rotation; internal/external rotation
Webster et al. [55]	PT ^b			
	Participants: 18 (17 M, 1F) Age: 23.8 (6) Time between injury and surgery: 12.7 (11.6) wk Time since surgery: 10.9 (2) mo Surgical procedure: PT			
	Participants: 16 (13 M, 3 F) Age: 26 (6) Weight: 82 (11) Time between injury and surgery: NR Time since surgery: 3.3 (0.4) yr Surgical procedure: 2 PT, 14 HT	Contralateral limb as controls	Vicon motion analysis system, 50 Hz Kistler force plate	Walking: self-selected speed; footwear condition NR Kinematics: flexion; varus; internal rotation Kinetics: external moments flexion; extension; adduction

Table 3 continued

Study	ACLR group characteristics ^a	Control group characteristics ^a	Methodology of data collection	Task and measurable variable
Webster and Feller [17]	HT group^b Participants: 16 M Age: 27.5 (6) Weight: 79.5 (7) Time between injury and surgery: 10.7 (9) wk Time since surgery: 9.4 (3) mo Surgical procedure: HT PT group^b Participants: 16 M Age: 23.8 (6) Weight: 79.2 (6) Time between injury and surgery: 11.9 (11) wk Time since surgery: 11.2 (2) mo Surgical procedure: PT	Participants: 16 M Age: 25.0 (5) Weight: 75.9 (12)	Vicon motion analysis system Kistler force plate	Walking: self-selected speed; barefoot Kinematics: peak adduction angle Kinetics: external peak adduction moment; vertical ground reaction force
Zabala et al. [51]	Participants: 45 (26 M, 19 F) Age: 29.5 (6.1) Weight: 74.4 (12.4) Time between injury and surgery: 2.2 (0.4–8.0) mo Time since surgery: 26 months (22–36) Surgical procedure: Achilles tendon	Participants: 45 (26 M, 19 F) Age: 30.2 (4.68) Weight: 74.2 (12.3)	Optoelectronic motion capture system, 120 Hz Force plate	Walking: self-selected speed, footwear condition NR Kinetics: (1st and 2nd peak) adduction external moments; abduction/adduction; flexion/extension; external/internal rotation moments Stair ascent/descent Kinetics: (1st and 2nd peak) adduction external moments; abduction/adduction; flexion/extension; external/internal rotation moment

ACL anterior cruciate ligament, ACLR anterior cruciate ligament reconstruction, AP anterior posterior, AMP anteromedial portal, BPTB bone patellar tendon bone, BTB bone patellar tendon bone, CKC close kinetic chain, CTO contralateral toe off, CFS contralateral foot strike, DB double bundle, DGA dimensional gait analysis, Fz vertical forces, Fx horizontal antero-posterior forces, Fy medio-lateral forces, F female, FS foot strike, HS heel strike, HT hamstring tendon, IE internal-external, LAD ligament augmentation device, L5 lumbar 5 vertebral level, M male, ML medio-lateral, MS mid-stance, Max maximum, NR not reported, OKC open kinetic chain, PT patellar tendon, ROM range of motion, SI sacrum 1 vertebral level, SB single bundle, SD standard deviation, ST semitendinosus, TO toe off, TT trans-tibial, VV varus-valgus, yr year(s)

^a Age is presented in years as mean (SD); weight is presented in kg

^b Groups combined for meta-analysis

^c Both groups considered separately for meta-analysis

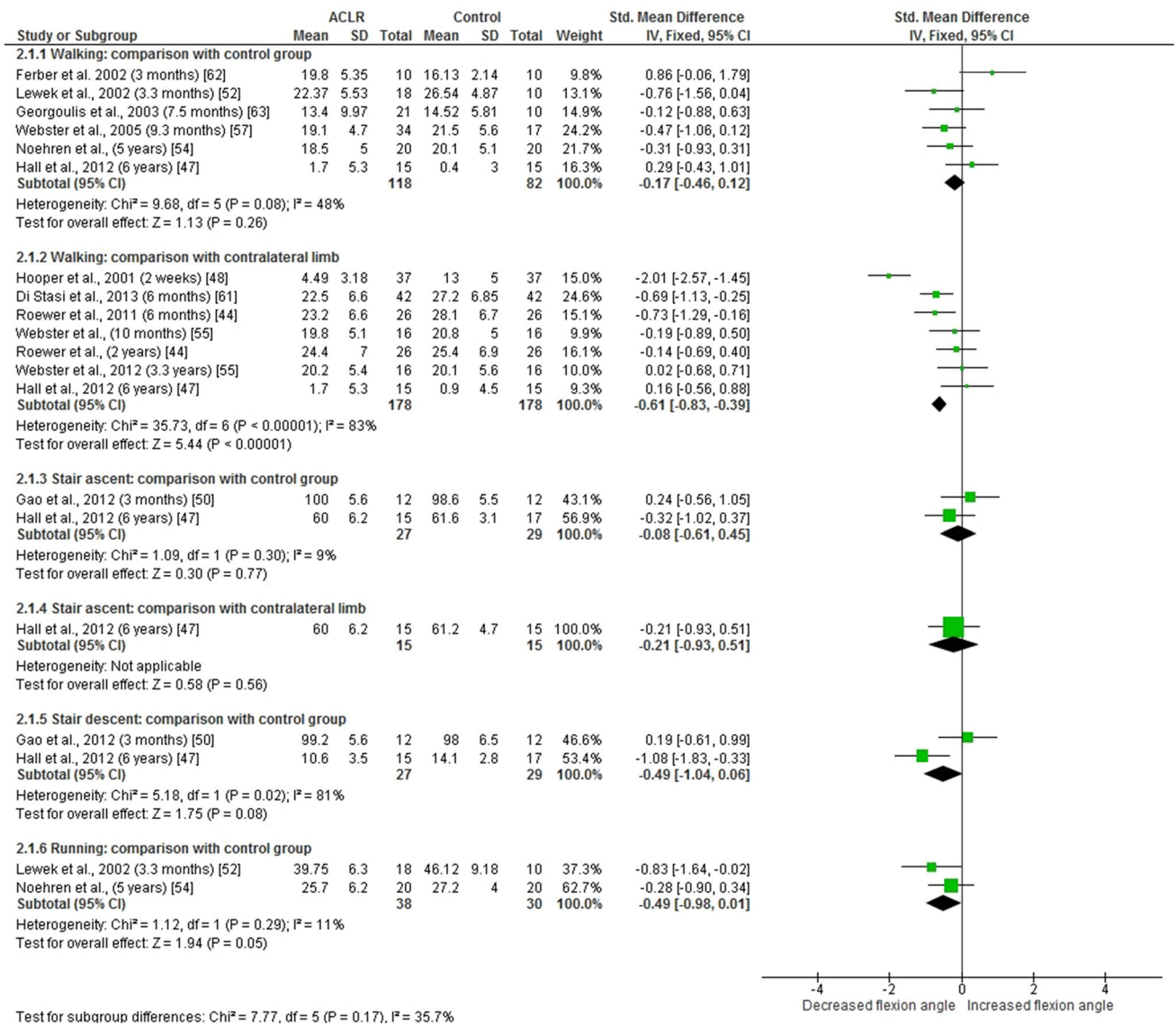


Fig. 2 Forest plot for peak knee flexion angle during stance phase of gait during different activities. ACLR anterior cruciate ligament reconstruction, CI confidence interval, IV inverse variance, SD standard deviation

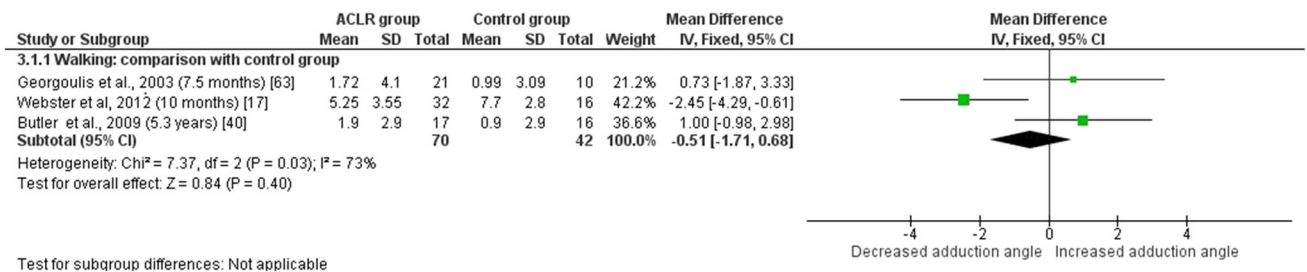


Fig. 3 Forest plot for peak knee adduction angle during stance phase of gait during walking compared with control group. ACLR anterior cruciate ligament reconstruction, CI confidence interval, IV inverse variance, SD standard deviation

reconstruction with the contralateral limbs. With the exclusion of that study, a significant difference was still evident for peak knee flexion angle between injured and

uninjured contralateral limb (ES -0.4 ; 95 % CI -0.60 to -0.12). No significant differences were found for peak knee flexion during stair ascent compared with controls

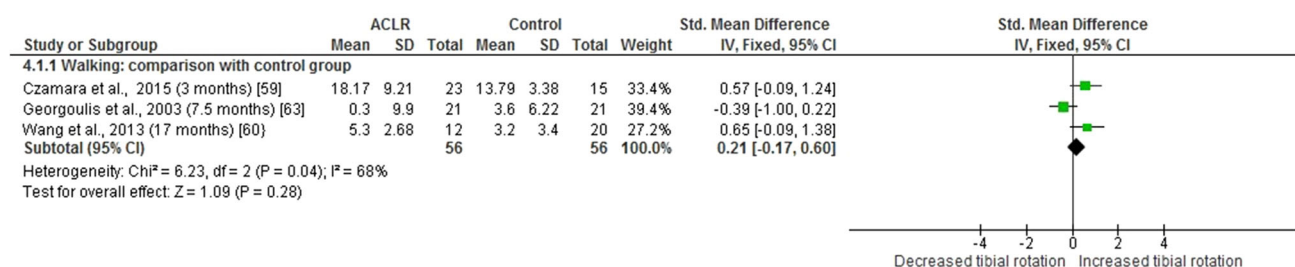


Fig. 4 Forest plot for peak tibial external rotation angle during stance phase of gait during walking compared with control group. *ACLR* anterior cruciate ligament reconstruction, *CI* confidence interval, *IV* inverse variance, *SD* standard deviation

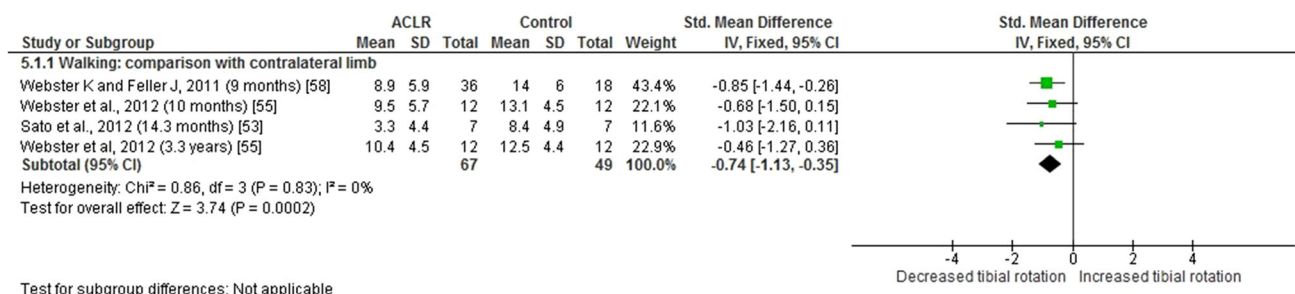


Fig. 5 Forest plot for peak tibial internal rotation angle during stance phase of gait compared with control group. *ACLR* anterior cruciate ligament reconstruction, *CI* confidence interval, *IV* inverse variance, *SD* standard deviation

(Fig. 2 [2.1.3]) and the contralateral limbs [47]. However, there was moderate evidence that peak flexion was decreased for ACLR knees compared with controls during stair descent (Fig. 2 [2.1.5]). During running, no significant differences were found for the knee flexion angle compared with controls (Fig. 2 [2.1.6]).

Only three studies reported peak knee adduction angles (Fig. 3), with a range from 7.5 months to 5.3 years post-reconstruction [17, 40, 63]. The meta-analysis indicated moderate evidence for no significant difference between the ACLR knee and controls.

In terms of rotation at the knee, studies either reported maximal external or internal tibial rotation during the loading phase [53, 58–60, 63]. Strong evidence indicated no significant difference for external tibial rotation for ACLR knees compared with controls during walking (maximum 17 months post-surgery, Fig. 4). While peak external rotation angles were considered for analysis, Czamara et al. [59] reported external rotation at initial foot contact. Their results were included into the meta-analysis, as peak external rotation is most likely to occur at heel strike phase during the stance phase of the walking gait [64]. One study [50] found less external rotation for the ACLR knees at 3 months post-surgery during stair descent (ES -1.1 ; 95 % CI -2 to -0.2 ; MD -5.8° ; 95 % CI -1.8 to -9.9). Strong evidence was found for less tibial internal rotation (ES -0.7 ; 95 % CI

-1.1 to -0.4) (Fig. 5) for ACLR knees compared with the contralateral limbs during walking [53, 55, 58]. However, results of one study [41] indicated no significant difference between the ACLR and the contralateral limbs while running, with limited evidence (ES -1.1 ; 95 % CI -2.3 to 0).

3.4.2 Joint Moments

Overall, participants with ACLR presented with significantly lower peak flexion moments during walking than controls and the contralateral limbs (strong evidence) (Fig. 6 [6.2.1, 6.2.2]). Strong evidence was found for peak flexion moments being lower for the injured side than for the contralateral uninjured sides during walking, stair ascent, and descent after ACLR (Fig. 6 [6.2.3–6.2.6]). Moderate evidence indicated no significant differences during running in ACLR participants compared with controls (Fig. 6 [6.2.7]).

The meta-analysis showed moderate to strong evidence for significantly lower first peak adduction moments during walking for the ACLR knees when compared with controls and with the contralateral limbs, respectively (Fig. 7 [7.1.1, 7.1.2]). Strong and moderate evidence indicated no significant differences were present for the peak adduction moments during stair ascent and descent, respectively, for the ACLR knees compared with the control group (Fig. 7

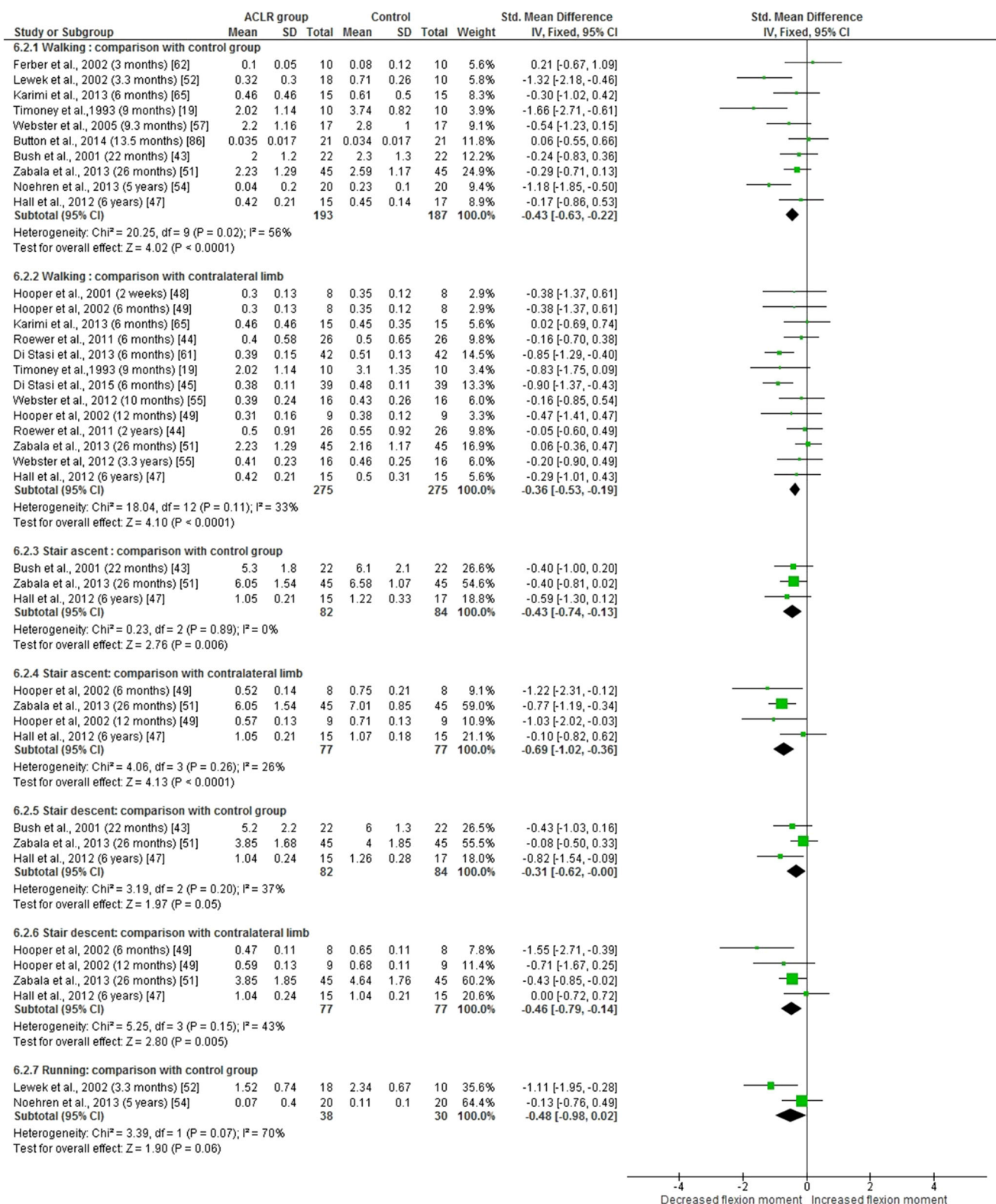


Fig. 6 Forest plot for knee flexion moment during stance phase of gait in different activities. ACLR anterior cruciate ligament reconstruction, CI confidence interval, IV inverse variance, SD standard deviation

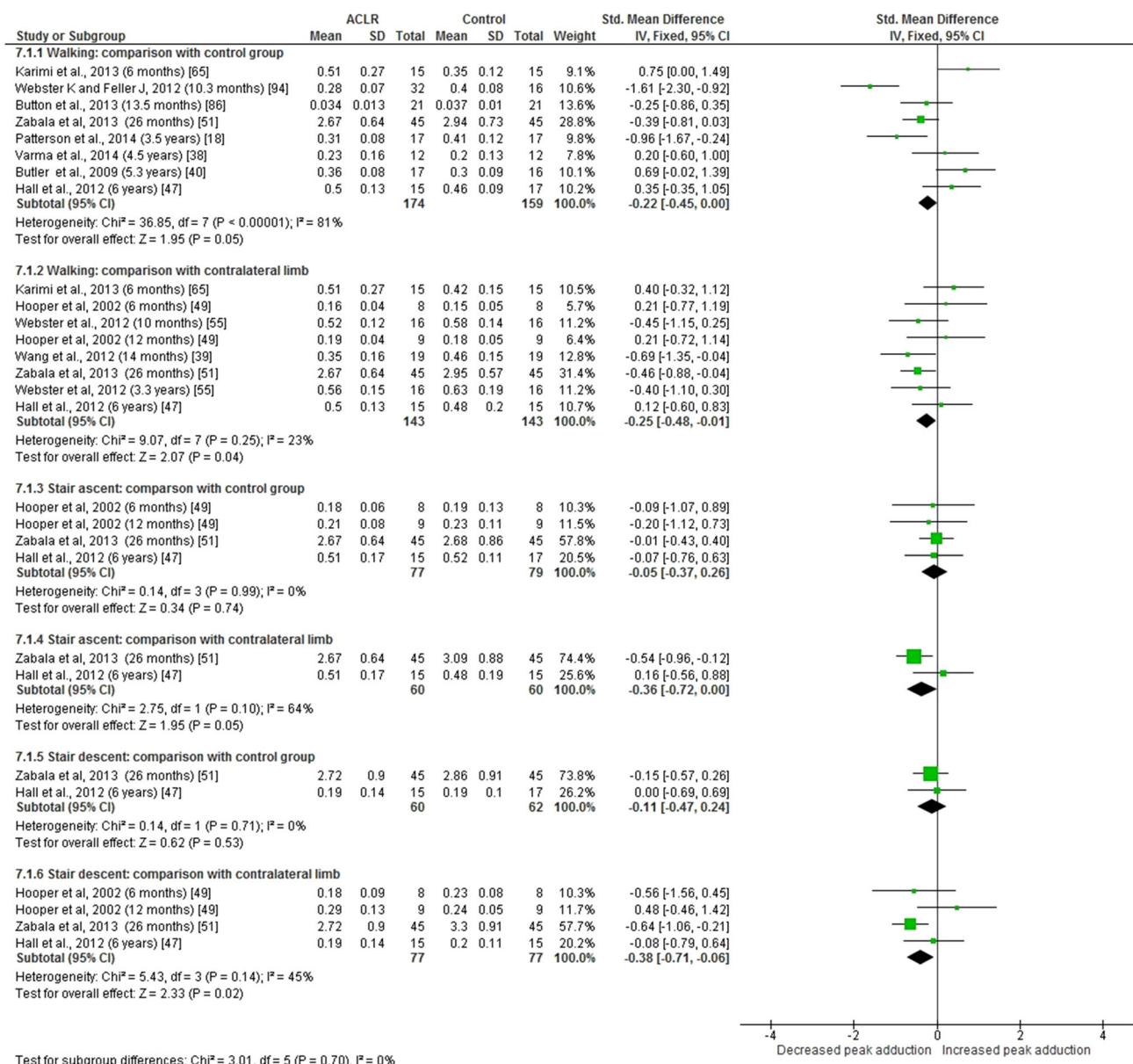


Fig. 7 Forest plot for first peak of knee adduction moment during stance phase of gait in different activities. ACLR anterior cruciate ligament reconstruction, CI confidence interval, IV inverse variance, SD standard deviation

[7.1.3, 7.1.5]). In contrast, lower peak adduction moments were found for ACLR during stair ascent (moderate evidence) and descent (strong evidence) compared with the contralateral limbs (Fig. 7 [7.1.4, 7.1.6]). No studies evaluating adduction moments during running were identified and included in the review.

ES for differences for the flexion and adduction moments were low (<0.59) for all comparisons, except for the comparison of flexion moments during stair ascent (ES 0.7, 95 % CI 0.1 to 0.4) and descent (ES 0.5, 95 % CI 0.1

to 0.8) when comparing ACLR knees with the contralateral limbs.

3.5 Time Course of Recovery

The qualitative assessment of the forest plots suggests patients tend to restore peak knee flexion angle over time after surgery for studies that included participants up to 6 years, and present peak knee flexion angle that is similar to the contralateral limb (Fig. 2). However, such an

Table 4 Effect sizes for joint angles between participants with anterior cruciate ligament reconstruction for between- and within-group comparisons

Variable	Task	Comparison	Level of evidence	Studies	ES (95 % CI)
Flexion angle	Walking	Controls	Strong	Four LR [47, 54, 57, 63], two HR [52, 62] studies; $I^2 = 48\%$, $p = 0.08$	-0.2 (-0.5 to 0.1)
		Contralateral limbs	Moderate	Four LR [44, 47, 55, 61], one HR [48] studies; $I^2 = 83\%$, $p < 0.00001$	-0.6 (-0.8 to -0.4)
	Stair ascent	Controls	Moderate	Two LR [47, 50] studies; $I^2 = 8\%$, $p = 0.30$	-0.0 (-0.6 to 0.5)
		Contralateral limbs	Limited	One LR [47] study	-0.2 (-0.98 to 0.5)
	Stair descent	Controls	Moderate	Two LR [47, 50] studies; $I^2 = 81\%$, $p = 0.02$	-0.5 (-1.0 to 0.0)
		Running	Moderate	One LR [54], one HR [52] studies; $I^2 = 11\%$, $p = 0.29$	-0.5 (-0.9 to 0.0)
Adduction angle	Walking	Controls	Moderate	Three LR [17, 40, 63] studies; $I^2 = 73\%$, $p = 0.03$	-0.5 (-1.7 to 0.7)
Tibial external rotation angle	Walking	Controls	Strong	Three LR [59, 60, 63] studies; $I^2 = 68\%$, $p = 0.04$	0.2 (-0.2 to 0.6)
	Stair ascent	Controls	Limited	One LR [50] study	-0.7 (-1.5 to 0.2)
	Stair descent	Controls	Limited	One LR [50] study	-1.1 (-2 to -0.2)
Tibial internal rotation angle	Walking	Contralateral limbs	Strong	Three LR [53, 55, 58] studies; $I^2 = 0\%$, $p = 0.63$	-0.7 (-1.1 to -0.4)
	Stair ascent	Controls	Limited	One LR [50] study	0.6 (-0.3 to 1.4)
	Stair descent	Controls	Limited	One LR [50] study	0.7 (-0.2 to 1.5)
	Running	Contralateral limbs	Limited	One LR [53] study	-1.1 (-2.3 to 0)

CI confidence interval, ES effect size, HR high risk of bias, LR low risk of bias

improvement over time was not evident for peak flexion moments during walking. When ascending and descending stairs, patients tended to present improved flexion moments over time when compared with the contralateral sides, but not when compared with controls (Fig. 6).

During walking, peak adduction moments appeared to be significantly lower in participants with ACLR than in controls within the first year following surgery (average 9.4 months) [70]. However, while not significant (95 % CI -0.02 to 1.4), these adduction moments showed a tendency to be slightly higher than in controls at a later phase (5.3 years) [40] (Fig. 7). Interestingly, Karimi et al. [65] found higher peak adduction moments at 6 months post-reconstruction than for controls. Functional braces were worn by their participants for 3 months post-surgery; none of the other studies reported these being worn. A sensitivity test was performed by excluding the adduction moments for this study [65], and the results indicate that, overall, peak adduction moments were significantly lower in the ACLR group than in the control group (ES -0.32, 95 % CI -0.55 to -0.08). When excluding their results, there appears to be a response in time, from lower to equivalent adduction moments for the ACLR group compared with controls while walking.

4 Discussion

The primary aim of this review was to investigate knee joint angles and moments in participants with ACLR compared with the contralateral limbs and with age-matched controls during walking, stair climbing, and running. In terms of joint angles, moderate to strong evidence was found for no differences for peak flexion during walking, stair ascent, and jogging when comparing the ACLR knees with controls. However, when compared with the uninjured contralateral sides, less flexion and less internal rotation was found for the ACLR knees during walking. During stair descent, peak flexion was decreased for ACLR knees compared with controls. There was moderate evidence for no differences for peak adduction angles between the ACLR knee and controls. Moderate to strong evidence was found for lower peak flexion moments in participants with ACLR during walking and stair ascent compared with controls and with the contralateral limbs. Moderate to strong evidence also indicated lower peak adduction moment for the ACLR knees during walking and stair ascent and descent than for the contralateral limbs, and during walking compared with the controls.

Table 5 Effect sizes for moments between participants with anterior cruciate ligament reconstruction for between- and within-group comparisons

Variable	Task	Comparison	Level of evidence	Studies	ES (95 % CI)
Flexion moments	Walking	Controls	Strong	Seven LR [43, 47, 51, 54, 57, 65, 86], three HR [19, 52, 62] studies, $I^2 = 56\%$, $p = 0.02$	-0.4 (-0.6 to -0.2)
		Contralateral limbs	Strong	Six LR [44, 45, 47, 51, 55, 61, 65], three HR [19, 48, 49] studies; $I^2 = 32\%$, $p = 0.13$	-0.4 (-0.5 to 0.2)
	Stair ascent	Controls	Strong	Three LR [43, 47, 51] studies, $I^2 = 0\%$, $p = 0.89$	-0.4 (-0.7 to -0.1)
		Contralateral limbs	Strong	Two LR [47, 51], one HR [49] studies, $I^2 = 26\%$, $p = 0.26$	-0.7 (-1.0 to -0.4)
	Stair descent	Controls	Strong	Three LR [43, 47, 51] studies, $I^2 = 37\%$, $p = 0.20$	-0.3 (-0.7 to 0)
		Contralateral limbs	Strong	Two LR [47, 51]; one HR [49] studies, $I^2 = 43\%$, $p = 0.15$	-0.5 (-0.8 to -0.1)
	Running	Controls	Limited	One LR [54], one HR [52] studies, $I^2 = 70\%$, $p = 0.07$	-0.5 (-0.9 to 0.2)
		Contralateral limbs	Limited	One LR [54], one HR [52] studies, $I^2 = 70\%$, $p = 0.07$	-0.5 (-0.9 to 0.2)
Adduction moments	Walking	Controls	Strong	Eight LR [18, 38, 40, 47, 51, 65, 86, 94] studies, $I^2 = 81\%$, $p < 0.0001$	-0.2 (-0.5 to 0)
		Contralateral limbs	Strong	Five LR [39, 47, 51, 55, 60, 65], one HR [49] studies; $I^2 = 0\%$, $p = 0.47$	-0.3 (-0.6 to -0.1)
	Stair ascent	Controls	Moderate	Two LR [47, 51], one HR [49] studies; $I^2 = 0\%$, $p = 0.99$	-0.1 (-0.4 to 0.3)
		Contralateral limbs	Strong	Two LR [47, 51] studies; $I^2 = 64\%$, $p = 0.10$	-0.4 (-0.7 to 0)
	Stair descent	Controls	Strong	Two LR [47, 51] studies; $I^2 = 0\%$, $p = 0.71$	-0.1 (-0.5 to 0.2)
		Contralateral limbs	Moderate	Two LR [47, 51], one HR [49] studies; $I^2 = 45\%$, $p = 0.14$	-0.4 (-0.7 to -0.1)
	Running	Controls	Limited	One LR [54], one HR [52] studies, $I^2 = 70\%$, $p = 0.07$	-0.5 (-0.9 to 0.2)
		Contralateral limbs	Limited	One LR [54], one HR [52] studies, $I^2 = 70\%$, $p = 0.07$	-0.5 (-0.9 to 0.2)

CI confidence interval, ES effect size, HR high risk of bias, LR low risk of bias

4.1 Peak Joint Angles

Results of the meta-analysis suggest that participants with ACLR regain similar peak knee flexion angles compared with the control group during walking by 6 years following surgery (Fig. 2). However, pooled peak flexion angles were lower in the ACLR knee than in the contralateral sides (ES 0.6; 95 % CI -0.4 to -0.8). The MD of 4.3° was higher than the minimal clinically important difference of 3° reported for knee flexion during the stance phase of walking [66]. This difference was influenced in part by the findings by Hooper et al. [48] at 2 weeks post-surgery: the MD was 8.5°, which may be due to pain or swelling at early stages after surgery. Considering the individual studies, limb asymmetry for peak flexion angles was evident until 6 months post-reconstruction [44, 61], while no significant bilateral differences were found, on average, 10 months to 6 years following surgery.

Similarly, our meta-analysis indicated no significant between-group differences for knee adduction and external

rotation angles during walking. However, peak external rotation was decreased during stair descent based on one study [50]. Internal rotation was decreased during walking for the ACLR knee compared with the contralateral limb. Rotational shifts varied from 1 to 5.8°. A biomechanical model of degeneration showed that a 5° rotational shift may be enough to cause accelerated degeneration of the cartilage [67] due to alterations in the stresses applied to joint surfaces [68]. Whether these small differences in kinematics for participants with ACLR compared with controls and contralateral limbs indicate an increased risk for PTOA still needs confirmation.

4.2 Joint Moments

Strong evidence was found that peak flexion moments were significantly lower in participants with ACLR during walking and stair ascent and descent activities compared with controls and contralateral limbs. However, the ESs

were small during walking and when compared with the controls during stair ascent and descent, while moderate ESs were found for decreased flexion moments when compared with the contralateral sides during stair ascent and descent. It is thus possible that an asymmetry may be more evident during stair ascent and descent than during walking. Peak external flexion moments are considered to reflect net quadriceps function [69], and reduced moments may be associated with lower quadriceps strength [22]. Neurophysiological changes following ACL injury and reconstruction have been shown to include decreased lower limb muscle activation, as determined by electromyography in ACLR knees [70] and lower cortical activation in injured limbs compared with an uninjured and healthy population, contributing towards reduced net quadriceps function [71]. The presence of pain, reported even at 2–5 years after surgery [72], or reflex inhibition [73], could further influence the movement patterns, as apparent with reduced flexion moments. Furthermore, compensatory movements at the hip and trunk may also decrease knee joint loading [74]. It is thus likely that multiple mechanisms contribute towards long-term decreased flexion moments in the ACLR knees.

Similar to external knee flexion moments, the adduction moments for the ACLR knees were significantly smaller during walking than for controls, albeit with low ESs. No significant differences were found for either the flexion moments or the adduction moments during stair descent for the ACLR knees compared with the controls, while they were lower for the ACLR knees than for the contralateral side. This indicates a bilateral asymmetry during this task and could infer that, during stair descent, the contralateral side has higher flexion and adduction moments than controls. While this is speculative, compensatory patterns may be evident for the contralateral side [25], which are likely due to central nervous system changes following the injury and surgery. Such changes have been documented in patients with ACL-deficient knees [74].

The meta-analysis along with sensitivity test for one study [65] showed peak adduction moments were lower in ACLR participants than in controls in the short and medium term (up to 3.5 years) and higher as time following reconstruction progresses. Our findings contrast thus with those by Hart et al. [13], who recently found that the frontal plane moments may not be affected by ACLR. Thus, our findings suggest that frontal plane moments cannot yet be disregarded as outcomes and long-term consequences of ACLR. The finding of lower adduction moments in the ACLR knees contrast with increased adduction moments often reported in patients with knee OA [83]. Risk of progression of OA in patients with that disorder increases 6.46 times with a 1 % increase in adduction moments [15], hence the magnitude of adduction moments is correlated

with OA severity [12, 16]. While decreased adduction moments in the early phase following ACLR may reflect decreased weight-bearing and compensatory strategies of the trunk, hip, and ankle [83–85], a combination of changing flexion and adduction moments and joint rotations may contribute towards loading at unconditioned areas of the cartilage, potentially leading to development of OA [75]. Studies included in the present review did not report radiographic OA changes for their participants; therefore, whether those with higher adduction moments present with radiographic signs of OA is unknown [15]. Whether changes for external adduction moments during functional tasks predict onset of OA following ACLR still needs to be substantiated. While the development of OA is multi-factorial, these biomechanical factors should be explored longitudinally in all planes.

4.3 Clinical Implications

Overall, reduced flexion and adduction moments indicate altered joint loading in the long term following ACLR. These changes are likely associated with quadriceps weakness [76] and various neurophysiological mechanisms [77–79], and are potentially associated with risk of early-onset joint degeneration [80, 81] due to suboptimal joint loading during functional tasks such as walking, running, and stair navigation [82]. Altered kinematics, along with reduced flexion moments, may significantly change the stress distribution within the cartilage and initiate a degenerative process [83]. Di Stasi et al. [61] showed that athletes with superior functional performance at 6 months following ACLR appear to demonstrate less asymmetrical gait patterns than those that did not pass functional ‘return to play’ criteria. Therefore, it appears to be important to include assessment of potential gait asymmetries following ACLR in decision making for readiness for rehabilitation progression and return to sports [61, 84]. While small differences in walking may not be detectable on clinical examination, these may be more evident during high knee-loading activities, such as stair ascent/descent, running [84], and jumping [85]. The finding of movement and loading asymmetries in the current review indicate that it may be important for people with ACLR to continue with exercise programs focusing on strength, neuromuscular control, and proprioception even after the rehabilitation period is completed.

4.4 Methodological Considerations and Directions for Future Research

The risk-of-bias assessment identified methodological limitations for the included studies: the absence of outcome measurer blinding, reporting reliability of methodology,

lack of confounding factors such as control of gait speed, and a priori power calculation for sample size. While blinding of the researcher is difficult during data collection due to visible scarring at the operated knee, blinding for the participant group (control vs. ACLR) can be maintained while processing kinematic and kinetic data.

A total of 16 studies included in the meta-analysis controlled the speed of the tasks [17, 19, 38, 40, 41, 43–45, 52, 54, 57, 60–63, 86]. A compensatory strategy to maintain or decrease joint loading is to walk slower [87]. Thus, in those studies where speed was not reported, it is possible that participants with ACLR may have walked slower, thereby maintaining or controlling the joint loading effectively. This could lead to no significant differences between groups, as seen for stair descent for flexion and adduction moments. Furthermore, other compensatory strategies that develop over time post-reconstruction, such as transferring the load to the hip and ankle [88] and compensation with trunk lean, can also be used to decrease joint moments, as shown in patients with knee OA [89, 90]. These factors should be addressed in future research.

Only cross-sectional studies or baseline measures of RCTs were considered for the purpose of time course of recovery analysis in this review. However, longitudinal studies following patients at different time points following ACLR are needed to fully investigate how gait biomechanics change over time, and how these changes may influence the development of OA or re-injury. Furthermore, the aim of this review was to explore differences for peak knee joint angles and moments when comparing people with ACLR versus controls, and not between different surgical or rehabilitative programs. Findings of groups of participants with ACLR undergoing different procedures within individual studies were thus combined into grand means for the meta-analyses, which masks differences between these groups. Findings of this review are thus limited to comparisons with controls and when comparing the ACLR knees to the contralateral uninjured sides, irrespective of surgical methods. Finally, this review was limited to the discrete variables of peak angles and moments, and did not explore biomechanical variables in other phases of the gait cycle.

5 Conclusion

Joint kinematics of ACLR knees were found to be similar to control groups during walking and stair navigation within a few months after surgery. However, differences in knee external moments persist over a longer period. The meta-analysis indicated lower pooled external flexion moments for people with ACLR than for controls during walking and stair ascent. The meta-analysis also indicated

decreased peak adduction moments during walking and stair navigation for ACLR knees, specifically when compared with contralateral limbs. Based on individual studies included in the review, such moments may be lower than for controls in the early phase following ACLR (10 months), but higher at a later phase (5 years) after surgery. While this review did not explore compensatory mechanisms, it is likely that multiple mechanisms influence altered gait patterns, including changes in muscle function and neuromuscular control at the trunk, hip, knee, and ankle. Gait patterns at the knee are not fully restored after more than 5 years following surgery. This finding indicates that long-term rehabilitative or maintenance programs may need to be considered for patients undergoing ACLR.

Author contributions All authors contributed significantly to the review. Gisela Sole was responsible for the conception of this review; Mandeep Kaur conducted the entire review, with Gisela Sole and Daniel Cury Ribeiro as second and third reviewers; Jean-Claude Theis and Kate Webster provided advice throughout the review. All authors were involved with the final manuscript.

Compliance with Ethical Standards

Funding No sources of funding were used to assist in the preparation of this article.

Conflict of interest Mandeep Kaur, Daniel Cury Ribeiro, Jean-Claude Theis, Kate Webster, and Gisela Sole declare that they have no conflicts of interest relevant to the content of this review.

References

1. Thing LF. “Voices of the broken body”. The resumption of non-professional female players’ sports careers after anterior cruciate ligament injury. The female player’s dilemma: is she willing to run the risk? *Scand J Med Sci Sports*. 2006;16(5):364–75.
2. Gianotti SM, Marshall SW, Hume PA, et al. Incidence of anterior cruciate ligament injury and other knee ligament injuries: a national population-based study. *J Sci Med Sport*. 2009;12(6):622–7.
3. Bjordal JM, Arnøy F, Hannestad B, et al. Epidemiology of anterior cruciate ligament injuries in soccer. *Am J Sports Med*. 1997;25(3):341–5.
4. Beynnon BD, Johnson RJ, Abate JA, et al. Treatment of anterior cruciate ligament injuries, part I. *Am J Sports Med*. 2005;33(10):1579–602.
5. Collins JE, Katz JN, Donnell-Fink LA, et al. Cumulative incidence of ACL reconstruction after ACL injury in adults: role of age, sex, and race. *Am J Sports Med*. 2013;41(3):544–9.
6. Ardern CL, Taylor NF, Feller JA, et al. Fifty-five per cent return to competitive sport following anterior cruciate ligament reconstruction surgery: an updated systematic review and meta-analysis including aspects of physical functioning and contextual factors. *Br J Sports Med*. 2014;48(21):1543–52.
7. Lohmander LS, Englund PM, Dahl LL, et al. The long-term consequence of anterior cruciate ligament and meniscus injuries: osteoarthritis. *Am J Sports Med*. 2007;35(10):1756–69.

8. Smith T, Postle K, Penny F, et al. Is reconstruction the best management strategy for anterior cruciate ligament rupture? A systematic review and meta-analysis comparing anterior cruciate ligament reconstruction versus non-operative treatment. *Knee*. 2014;21(2):462–70.
9. Oiestad BE, Holm I, Aune AK, et al. Knee function and prevalence of knee osteoarthritis after anterior cruciate ligament reconstruction: a prospective study with 10 to 15 years of follow-up. *Am J Sports Med*. 2010;38(11):2201–10.
10. Keays SL, Newcombe PA, Bullock-Saxton JE, et al. Factors involved in the development of osteoarthritis after anterior cruciate ligament surgery. *Am J Sports Med*. 2010;38(3):455–63.
11. Nigg BM, MacIntosh BR, Mester J, et al. Biomechanics and biology of movement. 1st ed. Champaign: Human Kinetics; 2000.
12. Foroughi N, Smith R, Vanwanseele B, et al. The association of external knee adduction moment with biomechanical variables in osteoarthritis: a systematic review. *Knee*. 2009;16(5):303–9.
13. Hart HF, Culvenor AG, Collins NJ, et al. Knee kinematics and joint moments during gait following anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Br J Sports Med*. 2015. doi:10.1136/bjsports-2015-094797.
14. Schipplein O, Andriacchi T. Interaction between active and passive knee stabilizers during level walking. *J Orthop Res*. 1991;9(1):113–9.
15. Miyazaki T, Wada M, Kawahara H, et al. Dynamic load at baseline can predict radiographic disease progression in medial compartment knee osteoarthritis. *Ann Rheum Dis*. 2002;61(7):617–22.
16. Sharma L, Hurwitz DE, Eugene JAT, et al. Knee adduction moment, serum hyaluronan level, and disease severity in medial tibiofemoral osteoarthritis. *Arthritis Rheumatol*. 1998;6(9):1233–40.
17. Webster KE, Feller JA. The knee adduction moment in hamstring and patellar tendon anterior cruciate ligament reconstructed knees. *Knee Surg Sports Traumatol Arthrosc*. 2012;20(11):2214–9.
18. Patterson MR, Delahunt E, Caulfield B, et al. Peak knee adduction moment during gait in anterior cruciate ligament reconstructed females. *Clin Biomech*. 2014;29(2):138–42.
19. Timoney JM, Inman WS, Quesada PM, et al. Return of normal gait patterns after anterior cruciate ligament reconstruction. *Am J Sports Med*. 1993;21(6):887–9.
20. Bulgheroni P, Bulgheroni MV, Andriani L, et al. Gait patterns after anterior cruciate ligament reconstruction. *Knee Surg Sports Traumatol Arthrosc*. 1997;5(1):14–21.
21. Sosdian L, Dobson F, Wrigley T, et al. Longitudinal changes in knee kinematics and moments following knee arthroplasty: a systematic review. *Knee*. 2014;21(6):994–1008.
22. Schmitt LC, Paterno MV, Ford KR, et al. Strength asymmetry and landing mechanics at return to sport after anterior cruciate ligament reconstruction. *Med Sci Sports Exerc*. 2015;47(7):1426–34.
23. Andriacchi TP. Dynamics of pathological motion: applied to the anterior cruciate deficient knee. *J Biomech*. 1990;23(1):99–105.
24. Gokeler A, Benjaminse A, Van Eck CF, et al. Return of normal gait as an outcome measurement in ACL reconstructed patients. A systematic review. *Int J Sports Phys Ther*. 2013;8(4):441–51.
25. Hart JM, Ko WK, Konold T, et al. Sagittal plane knee joint moments following anterior cruciate ligament injury and reconstruction: a systematic review. *Clin Biomech*. 2010;25(4):277–83.
26. Kutzner I, Heinlein B, Graichen F, et al. Loading of the knee joint during activities of daily living measured in vivo in five subjects. *J Biomech*. 2010;43(11):2164–73.
27. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann Intern Med*. 2009;151(4):264–9.
28. Downs SH, Black N. The feasibility of creating a checklist for the assessment of the methodological quality both of randomised and non-randomised studies of health care interventions. *J Epidemiol Community Health*. 1998;52(6):377–84.
29. Mündermann A, Dyrby CO, Hurwitz DE, et al. Potential strategies to reduce medial compartment loading in patients with knee osteoarthritis of varying severity: reduced walking speed. *Arthritis Rheum*. 2004;50(4):1172–8.
30. Barton CJ, Lack S, Malliaras P, et al. Gluteal muscle activity and patellofemoral pain syndrome: a systematic review. *Br J Sports Med*. 2013;47(4):207–14.
31. Cohen J. Statistical power analysis for behavioural sciences. Hillsdale: Lawrence Erlbaum Associates. Inc.:[sn]; 1988.
32. Van Tulder M, Furlan A, Bombardier B, et al. Updated method guidelines for systematic reviews in the Cochrane Collaboration Back Review Group. *Spine*. 2003;28(12):1290–9.
33. Schmalz T, Blumentritt S, Wagner R, et al. Gait analysis of patients within one year after ACL reconstruction. *Phys Med Rehab Kuror*. 1998;8(1):1–8.
34. Schmalz T, Blumentritt S, Wagner R, et al. Evaluation with biomechanical gait analysis of various treatment methods after rupture of the anterior cruciate ligament. *Sportverletzung Sports*. 1998;12(4):131–7 (**German**).
35. Wang WM, Zhao DW, Cui DP, et al. Gait analysis associated with anterior cruciate ligament reconstruction. *Zhonghua yi Xue Za Zhi*. 2009;89(29):2025–9 (**Chinese**).
36. Scanlan SF, Chaudhari AM, Dyrby CO, et al. Differences in tibial rotation during walking in ACL reconstructed and healthy contralateral knees. *J Biomech*. 2010;43(9):1817–22.
37. Gao B, Zheng NN. Alterations in three-dimensional joint kinematics of anterior cruciate ligament-deficient and -reconstructed knees during walking. *Clin Biomech*. 2010;25(3):222–9.
38. Varma RK, Duffell LD, Nathwani D, et al. Knee moments of anterior cruciate ligament reconstructed and control participants during normal and inclined walking. *BMJ Open*. 2014;4(6):004753.
39. Wang H, Fleischli JE, NigelZheng N. Effect of lower limb dominance on knee joint kinematics after anterior cruciate ligament reconstruction. *Clin Biomech*. 2012;27(2):170–5.
40. Butler RJ, Minick KI, Ferber R, et al. Gait mechanics after ACL reconstruction: implications for the early onset of knee osteoarthritis. *Br J Sports Med*. 2009;43(5):366–70.
41. Kuenze C, Hertel J, Weltman A, et al. Jogging biomechanics after exercise in individuals with ACL-reconstructed knees. *Med Sci Sports Exerc*. 2014;46(6):1067–76.
42. Schroeder MJ, Krishnan C, Dhaer YY, et al. The influence of task complexity on knee joint kinetics following ACL reconstruction. *Clin Biomech*. 2015;30(8):852–9.
43. Bush-Joseph CA, Hurwitz DE, Patel RR, et al. Dynamic function after anterior cruciate ligament reconstruction with autologous patellar tendon. *Am J Sports Med*. 2001;29(1):36–41.
44. Roewer BD, Di Stasi SL, Snyder-Mackler L, et al. Quadriceps strength and weight acceptance strategies continue to improve two years after anterior cruciate ligament reconstruction. *J Biomech*. 2011;44(10):1948–53.
45. Di Stasi S, Hartigan EH, Snyder-Mackler L, et al. Sex-specific gait adaptations prior to and up to 6 months after anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 2015;45(3):207–14.
46. Kowalk DL, Duncan JA, McCue FC, et al. Anterior cruciate ligament reconstruction and joint dynamics during stair climbing. *Med Sci Sports Exerc*. 1997;29(11):1406–13.
47. Hall M, Stevermer CA, Gillette JC. Gait analysis post anterior cruciate ligament reconstruction: knee osteoarthritis perspective. *Gait Posture*. 2012;36(1):56–60.
48. Hooper DM, Morrissey MC, Drechsler W, et al. Open and closed kinetic chain exercises in the early period after anterior cruciate

- ligament reconstruction: improvements in level walking, stair ascent, and stair descent. *Am J Sports Med.* 2001;29(2):167–74.
49. Hooper DM, Morrissey MC, Drechsler WI, et al. Gait analysis 6 and 12 months after anterior cruciate ligament reconstruction surgery. *Clin Orthop Relat Res.* 2002;403:168–78.
 50. Gao B, Cordova ML, Zheng N, et al. Three-dimensional joint kinematics of ACL-deficient and ACL-reconstructed knees during stair ascent and descent. *Hum Mov Sci.* 2012;31(1):222–35.
 51. Zabala MA, Favre J, Scanlan SF, et al. Three-dimensional knee moments of ACL reconstructed and control subjects during gait, stair ascent, and stair descent. *J Biomech.* 2013;46(3):515–20.
 52. Lewek M, Rudolph K, Axe M, et al. The effect of insufficient quadriceps strength on gait after anterior cruciate ligament reconstruction. *Clin Biomech (Bristol, Avon).* 2002;17(1):56–63.
 53. Sato K, Maeda A, Takano Y, et al. Relationship between static anterior laxity using the KT-1000 and dynamic tibial rotation during motion in patients with anatomical anterior cruciate ligament reconstruction. *Kurume Med J.* 2013;60(1):1–6.
 54. Noehren B, Wilson H, Miller C, et al. Long-term gait deviations in anterior cruciate ligament-reconstructed females. *Med Sci Sports Exerc.* 2013;45(7):1340–7.
 55. Webster KE, Feller FA, Wittwer JE, et al. Longitudinal changes in knee joint biomechanics during level walking following anterior cruciate ligament reconstruction surgery. *Gait Posture.* 2012;36(2):167–71.
 56. EMathZone, CV. Reference for grand mean: EMathZone, Combined variance. Available from: <http://www.emathzone.com/tutorials/basic-statistics/combined-variance.html>. Accessed 7 Oct 2014.
 57. Webster KE, Wittwer JE, O'Brien J, et al. Gait patterns after anterior cruciate ligament reconstruction are related to graft type. *Am J Sports Med.* 2005;33(2):247–54.
 58. Webster KE, Feller JA. Alterations in joint kinematics during walking following hamstring and patellar tendon anterior cruciate ligament reconstruction surgery. *Clin Biomech (Bristol, Avon).* 2011;26(2):175–80.
 59. Czamara A, Markowska I, Królikowska A, et al. Kinematics of rotation in joints of the lower limbs and pelvis during gait: early results—SB ACLR approach versus DB ACLR approach. *Biomed Res Int.* 2015. doi:10.1155/2015/707168.
 60. Wang HS, Fleischli JE, Zheng NQ, et al. Transtibial versus anteromedial portal technique in single-bundle anterior cruciate ligament reconstruction outcomes of knee joint kinematics during walking. *Am J Sports Med.* 2013;41(8):1847–56.
 61. Di Stasi SL, Logerstedt D, Gardinier ES, et al. Gait patterns differ between ACL-reconstructed athletes who pass return-to-sport criteria and those who fail. *Am J Sports Med.* 2013;41(6):1310–8.
 62. Ferber R, Osternig LR, Woollacott MH, et al. Gait mechanics in chronic ACL deficiency and subsequent repair. *Clin Biomech.* 2002;17(4):274–85.
 63. Georgoulis AD, Papadonikolakis CD, Papageorgiou, et al. Three-dimensional tibiofemoral kinematics of the anterior cruciate ligament-deficient and reconstructed knee during walking. *Am J Sports Med.* 2003;31(1):75–9.
 64. LaFortune M, Cavanagh P, Sommer H, et al. Three-dimensional kinematics of the human knee during walking. *J Biomech.* 1992;25(4):347–57.
 65. Karimi M, Fatoye F, Mirbod SM, et al. Gait analysis of anterior cruciate ligament reconstructed subjects with a combined tendon obtained from hamstring and peroneus longus. *Knee.* 2013;20(6):526–31.
 66. Di Stasi SL, Snyder-Mackler L. The effects of neuromuscular training on the gait patterns of ACL-deficient men and women. *Clin Biomech.* 2012;27(4):360–5.
 67. Andriacchi TP, Mündermann A. The role of ambulatory mechanics in the initiation and progression of knee osteoarthritis. *Curr Opin Rheumatol.* 2006;18(5):514–8.
 68. Tashman S, Collon D, Anderson K, et al. Abnormal rotational knee motion during running after anterior cruciate ligament reconstruction. *Am J Sports Med.* 2004;32(4):975–83.
 69. Andriacchi TP, Dyrby CO. Interactions between kinematics and loading during walking for the normal and ACL deficient knee. *J Biomech.* 2005;38(2):293–8.
 70. Nyland J, Klein S, Caborn DN, et al. Lower extremity compensatory neuromuscular and biomechanical adaptations 2 to 11 years after anterior cruciate ligament reconstruction. *Arthroscopy.* 2010;26(9):1212–25.
 71. Kuenze CM, Hertel J, Weltman A, et al. Persistent neuromuscular and corticomotor quadriceps asymmetry after anterior cruciate ligament reconstruction. *J Athl Train.* 2015;50(3):303–12.
 72. Kartus J, Magnusson L, Stener S, et al. Complications following arthroscopic anterior cruciate ligament reconstruction: a 2- to 5-year follow-up of 604 patients with special emphasis on anterior knee pain. *Knee Surg Sports Traumatol Arthrosc.* 1999;7(1):2–8.
 73. Hodges PW, Tucker K. Moving differently in pain: a new theory to explain the adaptation to pain. *Pain.* 2011;152(3):S90–8.
 74. Courtney CA, Rine RM. Central somatosensory changes associated with improved dynamic balance in subjects with anterior cruciate ligament deficiency. *Gait Posture.* 2006;24(2):190–5.
 75. Andriacchi TP, Briant PL, Beville SL, et al. Rotational changes at the knee after ACL injury cause cartilage thinning. *Clin Orthop Relat Res.* 2006;442:39–44. doi:10.1097/01.blo.0000197079.26600.09.
 76. Shelburne KB, Torry MR, Pandy MG. Contributions of muscles, ligaments, and the ground-reaction force to tibiofemoral joint loading during normal gait. *J Orthop Res.* 2006;24(10):1983–90.
 77. Urbach D, Nebelung W, Röpke M, et al. Bilateral dysfunction of the quadriceps muscle after unilateral cruciate ligament rupture with concomitant injury central activation deficit. *Unfallchirurg.* 2000;103(11):949–55.
 78. Urbach D, Nebelung W, Becker R, et al. Effects of reconstruction of the anterior cruciate ligament on voluntary activation of quadriceps femoris a prospective twitch interpolation study. *J Bone Joint Surg Br.* 2001;83(8):1104–10.
 79. Barret DS. Proprioception and function after anterior cruciate reconstruction. *J Bone Joint Surg Br.* 1999;73(5):833–7.
 80. Roos H, Adalberth T, Dahlberg L, et al. Osteoarthritis of the knee after injury to the anterior cruciate ligament or meniscus: the influence of time and age. *Osteoarthritis Cartilage.* 1995;3(4):261–7.
 81. Roos PJ, Neu CP, Hull ML, et al. A new tibial coordinate system improves the precision of anterior-posterior knee laxity measurements: a cadaveric study using Roentgen stereophotogrammetric analysis. *J Orthop Res.* 2005;23(2):327–33.
 82. Knoll Z, Kiss RM, Kocsis L, et al. Gait adaptation in ACL deficient patients before and after anterior cruciate ligament reconstruction surgery. *J Electromyogr Kinesiol.* 2004;14(3):287–94.
 83. Setton LA, Elliott DM, Mow VC. Altered mechanics of cartilage with osteoarthritis: human osteoarthritis and an experimental model of joint degeneration. *Osteoarthritis Cartilage.* 1999;7(1):2–14.
 84. Myer GD, Chu DA, Brent JL, et al. Trunk and hip control neuromuscular training for the prevention of knee joint injury. *Clin Sports Med.* 2008;27(3):425–48.
 85. Paterno MV, Ford KR, Myer GD, et al. Limb asymmetries in landing and jumping 2 years following anterior cruciate ligament reconstruction. *Clin J Sport Med.* 2007;17(4):258–62.
 86. Button K, Roos PE, Van Deursen RW. Activity progression for anterior cruciate ligament injured individuals. *Clin Biomech (Bristol, Avon).* 2014;29(2):206–12.
 87. Robbins SM, Maly MR. The effect of gait speed on the knee adduction moment depends on waveform summary measures. *Gait Posture.* 2009;30(4):543–6.

88. Orishimo KF, Kremenik IJ, Mullaney MJ, et al. Adaptations in single-leg hop biomechanics following anterior cruciate ligament reconstruction. *Knee Surg Sports Traumatol Arthrosc.* 2010;18(11):1587–93.
89. Mündermann A, Dyrby CO, Andriacchi TP. Secondary gait changes in patients with medial compartment knee osteoarthritis: increased load at the ankle, knee, and hip during walking. *Arthritis Rheum.* 2005;52(9):2835–44.
90. Simic M, Hinman RS, Wrigley TV, et al. Gait modification strategies for altering medial knee joint load: a systematic review. *Arthritis Care Res.* 2011;63(3):405–26.
91. DeVita P, Hortobagyi T, Barrier J. Gait biomechanics are not normal after anterior cruciate ligament reconstruction and accelerated rehabilitation. *Med Sci Sports Exerc.* 1998;30(10):1481–8.
92. Ferber R, Osternig LR, Woollacott MH, et al. Bilateral accommodations to anterior cruciate ligament deficiency and surgery. *Clin Biomech.* 2004;19(2):136–44.
93. Tellini TL, Lima KO, Alouche SR, et al. Compliant surface after ACL reconstruction and its effects on gait. *Acta Sci Health Sci.* 2013;35(2):237–42.
94. Webster KE, Feller JA. Tibial rotation in anterior cruciate ligament reconstructed knees during single limb hop and drop landings. *Clin Biomech.* 2012;27(5):475–9.